

Wealth accumulation and capital mobility

The Solow model for a small open economy

In the basic Solow model the only source of domestic investment is the volume of domestic saving. This is a good approximation to reality as long as international capital flows are relatively small. However, in recent decades the international mobility of capital has increased dramatically. In these circumstances a country may finance part of domestic investment via capital imports, or it may invest part of domestic savings abroad through capital exports.

In this chapter we will develop a Solow model of an open economy to study how the dynamics of wealth accumulation are affected by international capital mobility. The rapid growth of international trade and capital mobility are two of the most striking features of the process of *globalization*. Most economists believe that capital mobility tends to improve the long-run efficiency of the world economy, but the liberalization of international capital flows also creates new risks. Indeed, many critics of globalization have argued that increased capital mobility may destabilize the world economy and expose poor countries to the shifting moods of international investors. The potential for capital flows to generate short-run macroeconomic instability will be discussed in Part 7, whereas the model developed in the present chapter will highlight the potential long-run benefits and risks associated with capital mobility. Studying this chapter along with those in Part 7 should thus give you a better understanding of some of the contentious issues in the current debate on globalization.

We will start the chapter by documenting the trend towards growing trade volumes and increasing capital mobility. We will then set up a Solow model of a small open economy whose domestic capital market is perfectly integrated with the world capital market. This model will be used to analyse how an open capital account¹ modifies the dynamics of wealth accumulation in the individual country, and how capital mobility and the associated opportunity to incur foreign debt brings benefits as well as potential costs.

1. Countries allowing free capital mobility are said to have an 'open capital account' since their governments do not control the items on the capital account of the balance of payments.

4.1 The growing mobility of capital

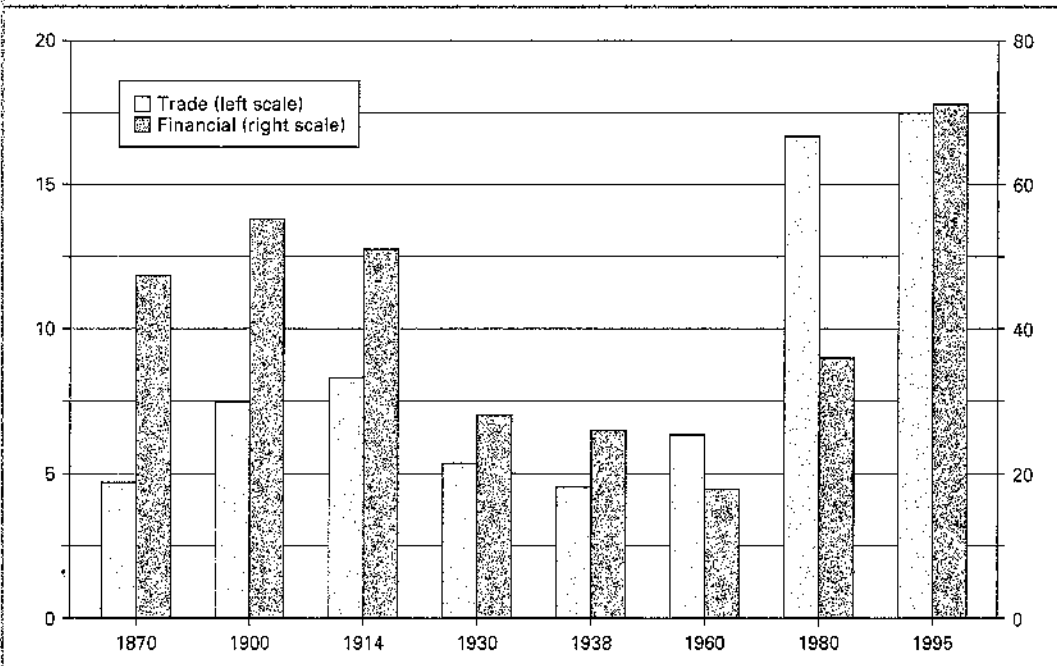
Figure 4.1 illustrates trends in the degree of international economic integration. In Figure 4.1a, reaching from the late nineteenth century until the end of the twentieth century, the light columns associated with the left scale measure the volume of international trade (the sum of exports and imports) relative to world GDP. The dark columns associated with the right scale show the ratio of the total stock of foreign-owned assets to world GDP. Thus the light columns indicate the degree of trade integration, and the dark columns illustrate the degree of financial integration (or capital mobility). The figure documents an interesting fact which is often forgotten in the contemporary debate on globalization: before the First World War international capital mobility was very high, especially considering that the technological means of communication were less developed than today. Indeed, the average ratio of foreign assets to GDP was above 50 per cent in the early twentieth century, as a result of large capital exports from the most developed countries (particularly Great Britain) to the less developed regions in the world. This is why critics of the economic dominance of the industrialized countries often refer to this period as the heyday of Western imperialism.

The First World War marked the beginning of a long period of economic and political instability, culminating in the Great Depression of the 1930s and in the catastrophe of the Second World War. During this dark period in history national governments retrenched into inward-looking economic policies involving strong trade protection and controls on capital flows. In Fig. 4.1a this is reflected in the sharp drop in trade integration and financial integration after 1914. Even as late as 1960 the world's economies were still considerably less open to trade and capital flows than they were before the First World War. In particular, there was much less financial integration in 1960 than in 1900, because most countries still maintained extensive restrictions on private capital flows across borders, in an attempt to maintain national autonomy in the conduct of monetary policy. However, between 1960 and 1980 the degree of financial integration roughly doubled, and from 1980 to 1995 it almost doubled once again, reaching the highest level ever recorded. To some extent this remarkable increase in capital mobility was driven by the simultaneous increase in world trade, since a part of international capital flows takes the form of trade credits granted by exporters to the foreign importers. But most of the increase in capital flows stemmed from a combination of financial liberalization – with one government after another abandoning capital controls – and the rapid improvements in information and communication technologies which made it easier and cheaper to move capital across borders.

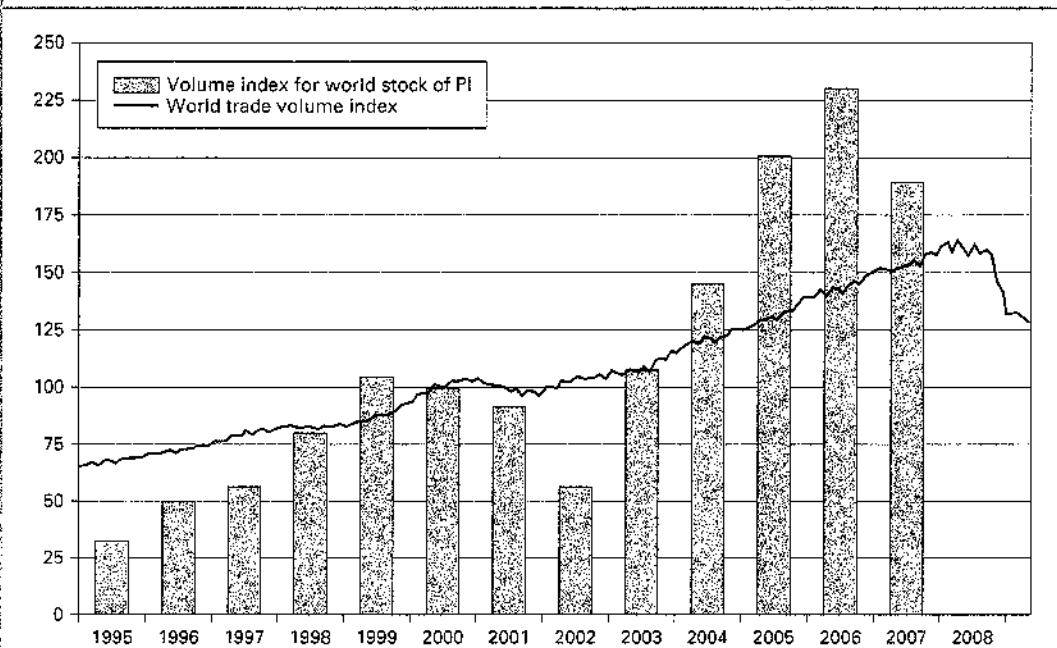
The increase in trade and financial integration has continued up to recently. Figure 4.1b shows how world trade (the sum of exports and imports for the countries of the world) and the stock of world portfolio investment have evolved since 1995. International portfolio investment involves international debt flows, e.g., the purchase of foreign bonds or the extension of bank credit across borders or the purchase of foreign shares for purposes of portfolio diversification. International portfolio investors do not acquire a controlling ownership share in foreign firms and therefore are not actively engaged in running a business firm abroad. This distinguishes portfolio investment, PI, from foreign direct investment, FDI. In Fig. 4.1b the stock of PI is used as measure of financial integration.

Between 1995 and 2007 world trade increased by 128 per cent, while the stock of PI increased by 511 per cent. In the same period world GDP rose by 58 per cent according to figures from the International Monetary Fund (IMF). Thus trade increased by more than twice as much as GDP and the stock of PI increased nine times the increase in GDP.

Figure 4.1b also shows the dramatic drop in world trade by more than 20 per cent from mid-2008 to mid-2009. This was a direct consequence of the great financial and economic crisis that exploded in the fall of 2008. This episode is, of course, of great interest theoretically and practically, but it is a business cycle episode and belongs to open economy macroeconomics for the short run dealt with in Part 7. Here our concern is with the long-run trend towards increased financial and trade integration.

FIGURE 4.1a Foreign trade and foreign assets relative to world GDP, 1870–1995

Source: Figure 3.3 of *World Economic Outlook*, International Monetary Fund, September 2002.

FIGURE 4.1b World trade volume index (monthly), and volume index for world stock of foreign portfolio investment (annual)

Source: CPB, Netherlands Bureau of Economic Analysis, *World Trade Monitor*, May 2009 and IMF, *Balance of Payments Statistics*.

TABLE 4.1 International investment, 1985–2002

	Stock of inward FDI % of GDP		Stock of outward FDI % of GDP		Stock of inward PI % of GDP		Stock of outward PI % of GDP	
	1985	2002	1985	2002	1986	2001	1986	2001
Australia	14.5	32.2	3.8	22.9	44.4	87.8	7.7	32.0
Austria	5.6	20.6	2.0	19.5	73.7	147.7	60.7	120.8
Canada	18.4	30.4	12.3	37.6	31.2	28.4	13.5	17.5
Denmark	6.0	41.7	3.0	43.4	—	120.3	—	89.6
Finland	2.5	27.0	3.4	52.8	30.7	153.8	22.9	81.2
France	6.9	28.2	7.1	45.8	2.8	74.7	2.8	58.7
Germany	5.1	22.7	8.4	29.0	20.6	62.9	11.7	60.0
Italy	4.5	10.6	3.9	16.4	19.9	62.5	13.3	56.3
Japan	0.3	1.5	3.2	8.3	10.9	18.0	14.6	33.3
Netherlands	18.8	74.9	36.1	84.7	71.0	181.2	78.5	220.1
Norway	11.7	17.4	1.7	20.0	54.8	—	23.3	—
Sweden	4.2	46.0	10.4	60.5	51.5	142.1	20.1	95.6
UK	14.1	40.8	22.0	66.7	40.4	116.2	51.3	112.1
USA	4.4	12.9	5.7	14.4	14.9	43.5	5.3	23.3
EU	9.3	31.4	10.5	41.0	—	—	—	—

Note: Definition of FDI: Investment in a lasting enterprise resident in another country. Definition of PI: Loans in foreign banks and cross-border investment in bonds, stocks, financial derivatives and other money market instruments.

Source: World Investment Report 2003; UNCTAD; IMF International Financial Statistics database.

As mentioned, international capital flows are often divided into foreign direct investment (FDI) and foreign portfolio investment (PI). FDI may take the form of 'greenfield' investment where a foreign multinational company sets up a new production plant in a country, but more often it involves the acquisition by a foreign company of a controlling ownership share in an existing domestic company. If an already established foreign-owned company retains and reinvests its profits in the host country, this is also counted as FDI. The distinguishing feature of FDI is that the investor is an active controlling owner who often brings along new production technologies and management know-how with the capital invested. Because FDI often implies an international transfer of technology and is typically undertaken by large multinational enterprises, it tends to attract special interest. Table 4.1 shows that FDI has become increasingly important relative to domestic economic activity in recent years. While FDI is a more visible form of investment, international portfolio investment (PI) still accounts for the bulk of international capital flows, as documented in Table 4.1.

In an open economy without restrictions on international capital flows, domestic saving, S_t , may either be invested in building up the domestic capital stock, K , or it may be used to build up foreign assets, F , via FDI or PI. In the open economy the intertemporal budget constraint of the domestic representative consumer (or of the domestic citizens together) is therefore given as:

$$F_{t+1} - F_t + K_{t+1} - K_t + \delta K_t = S_t, \quad (1)$$

where S_t is domestic gross saving in period t , $K_{t+1} - K_t + \delta K_t \equiv I_t$ is domestic gross investment in that period (which includes the replacement investment needed to compensate for the depreciation δK_t of the pre-existing capital stock), and $F_{t+1} - F_t$ is the rise in the stock of net foreign assets between period t and period $t + 1$. Thus:

$$F_{t+1} - F_t = S_t - I_t \quad (2)$$

is the amount of capital exported during period t . We see that capital exports are given by the difference between domestic saving and domestic investment: that part of saving which is not invested at home must be invested abroad. Of course, $S_t - I_t$ may be negative. In that case part of domestic investment is financed by capital imports from abroad.

One way of measuring the degree of international capital market integration is to calculate the correlation between national savings rates and national investment rates across countries. This may be done by estimating the parameters α_0 and α_1 in a regression equation of the following form where Y_t is real GDP in year t , and where the superscript i refers to country i :

$$(I_t/Y_t)^i = \alpha_0 + \alpha_1(S_t/Y_t)^i. \quad (3)$$

In a closed economy aggregate investment must equal aggregate saving, $I_t = S_t$, implying $\alpha_0 = 0$ and $\alpha_1 = 1$ in Eq. (3). In an open economy with free capital flows, investment and saving may deviate from each other, leading to capital export or import as shown by (2). Part of an increase in domestic saving will typically be invested abroad, implying that $\alpha_1 < 1$. The reason is that an increase in domestic saving tends to reduce the level of domestic real interest rates as the supply of capital goes up, and this fall in domestic interest rates induces savers to invest abroad rather than at home. The greater the degree of integration of the domestic and foreign capital markets, the greater will be the tendency for domestic savings to flow out of the country in response to any given drop in domestic interest rates, so the lower α_1 will tend to be.

Alternatively, one can measure the degree of international capital mobility by focusing directly on the imbalances on current accounts between countries. A country with a savings surplus, $S_t/Y_t > I_t/Y_t$, will have a surplus on the current account of its balance of payments, whereas a country with a savings deficit will run a current account deficit. To see this, recall that goods market equilibrium in an open economy requires

$$Y_t + M_t = C_t + I_t + X_t, \quad (4)$$

where C_t is total (private plus public) consumption, M_t is aggregate imports, and X_t is aggregate exports. If $\bar{r}$ is the international interest rate (assumed to be constant) and F_t is the stock of net foreign assets at the start of period t , then net asset income from abroad is $\bar{r}F_t$. We may rewrite (4) as:

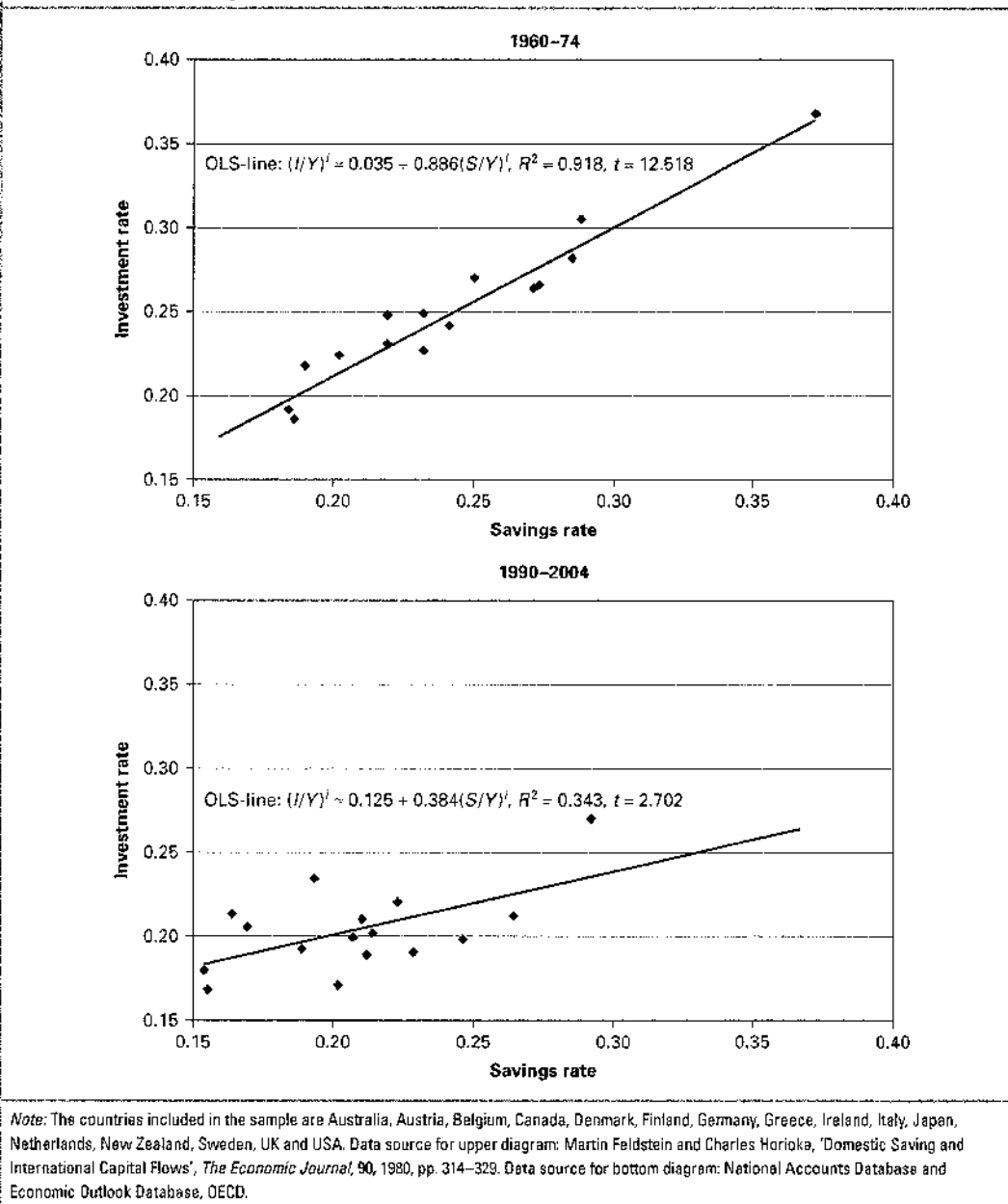
$$Y_t + \bar{r}F_t = C_t + I_t + X_t - M_t + \bar{r}F_t. \quad (5)$$

The magnitude $Y_t + \bar{r}F_t$ is total national income (the gross national product, GNP), consisting of gross domestic product, Y_t , plus net asset income from abroad, $\bar{r}F_t$. By definition, $X_t - M_t + \bar{r}F_t$ is the surplus on the current account of the balance of payments, consisting of the trade balance $X_t - M_t$ plus net foreign asset income, $\bar{r}F_t$. National saving is defined as national income minus total consumption, $S_t \equiv Y_t + \bar{r}F_t - C_t$. Denoting the current account balance by CA_t , we may therefore write (5) as:

$$CA_t = S_t - I_t = F_{t+1} - F_t, \quad CA_t \equiv X_t - M_t + \bar{r}F_t, \quad (6)$$

where (2) was used. This shows the well-known fact that the surplus on the current account and capital exports are just two sides of the same coin and both are equal to the savings surplus. The larger the average deviation between S_t and I_t across countries, the lower will be the estimated value of α_1 in (3). According to (6) a low value of α_1 therefore implies a large average imbalance on the current accounts of the countries considered.

Some years ago economists Martin Feldstein and Charles Horioka estimated a cross-country regression equation of the form (3), using average values of the investment rate,

FIGURE 4.2 National saving and national investment in 16 OECD countries

I_t/Y_t , and the savings rate, S_t/Y_t , for a sample of 16 OECD countries over the period 1960-74.² Their results are portrayed in the upper diagram in Fig. 4.2. The indicated line of best fit (estimated by OLS) has a slope as high as 0.89, suggesting a rather low degree of international capital mobility. The finding that savers apparently had a surprisingly high inclination to invest at home rather than abroad soon became known as 'the Feldstein-Horioka puzzle'. One possible explanation for the high correlation between domestic saving

2. Martin Feldstein and Charles Horioka, 'Domestic Saving and International Capital Flows', *Economic Journal*, 90, 1980, pp. 314-329.

TABEL 4.2 The correlation between national savings rate and national investment rate, 1975–2002 (estimates of α_1)

Period	Old OECD	European Union	Euro area
1975–2001	0.51	0.47	0.35
1975–1990	0.55	0.50	0.41
1991–2001	0.38	0.36	0.14

Source: Table 3 in Olivier Blanchard and Francesco Giavazzi, 'Current Account Deficits in the Euro Area: The End of the Feldstein–Horioka Puzzle?' *Brookings Papers on Economic Activity*, 2, 2002, pp. 147–209.

and domestic investment was that governments were eager to avoid large current account deficits because the OECD countries tried to maintain fixed exchange rates up until 1973. If a country started to build up a substantial current account deficit, its government would normally take steps to increase domestic saving and/or reduce domestic investment to redress its current account imbalance for fear that a large deficit would invite a speculative attack against its currency (by currency speculators believing that the country might have to devalue to rebalance its current account).

However, over the years the correlation between domestic saving and domestic investment has weakened. This is illustrated by the flatter regression line in the bottom part of Fig. 4.2, which shows a new estimation of Eq. (3) for the group of 16 OECD countries considered by Feldstein and Horioka, but using data for the period 1990–2004. We see that the slope coefficient α_1 for this recent period is less than one half the size of the original Feldstein–Horioka estimate, and the explanatory power of the regression as expressed by R -squared as well as the t -value is also considerably lower. Both of these findings suggest that international capital mobility has become significantly higher over time, thereby weakening the link between national saving and national investment.

Table 4.2 provides estimates of α_1 for three successive periods for a larger group of countries which are divided into the 'old' OECD (those OECD countries for which data are available for all the years back to 1975), the 15 countries which were members of the EU in 2001, and those EU countries which had adopted the euro in 2001. Again we see a clear tendency for α_1 to fall over time, indicating growing capital mobility. In particular, measured by the (low) value of α_1 , capital mobility in the euro area seems to have become very high. This was also reflected in the fact that a country like Portugal ran a current account deficit as large as 10 per cent of its GDP in 2000–01, up from 2–3 per cent at the start of the 1990s. With an investment rate I_t/Y_t of around 20 per cent, this meant that about half of all gross investment in Portugal was being financed by foreign investors at the turn of the century. Similarly, in Greece the current account deficit in 2000–01 was 6–7 per cent of GDP, up from 1–2 per cent in the early 1990s.

FACTS ABOUT INTERNATIONAL ECONOMIC INTEGRATION



The empirical evidence shows that since the 1960s the world's economies have become much more integrated both with respect to trade and with respect to capital flows. In particular, capital mobility between the richest countries of the world has increased dramatically and reached an unprecedentedly high level in 2008. The great financial crisis exploding in 2008 brought the intensification to a halt, but nevertheless the long-run tendency is towards dramatically increased integration, financially and with respect to trade.

In Chapter 3 we considered the benchmark case of an economy with no cross-border capital mobility at all. In the present chapter we consider the opposite benchmark case of

a small open economy whose capital market is perfectly integrated with the world capital market. The empirical evidence surveyed above suggests that capital mobility is neither absent nor perfect. Thus the 'truth' about capital mobility lies somewhere in between the two extreme cases studied in the previous and the current chapter, but looking at the two benchmark cases has the advantage of highlighting clearly what difference capital mobility can make for the dynamics of economic growth.

4.2 A Solow model of a small open economy

In this section we develop a growth model of a so-called small open economy with perfect mobility of commodities and capital, but with no labour mobility at all. This reflects the empirical fact that commodities and capital move much more across country borders than persons do. As examples of small open economies with free capital movements you may think of practically any of the countries in Western Europe, since even the larger economies like those of Britain, France or Germany are small compared to the size of the world economy.

The small open economy

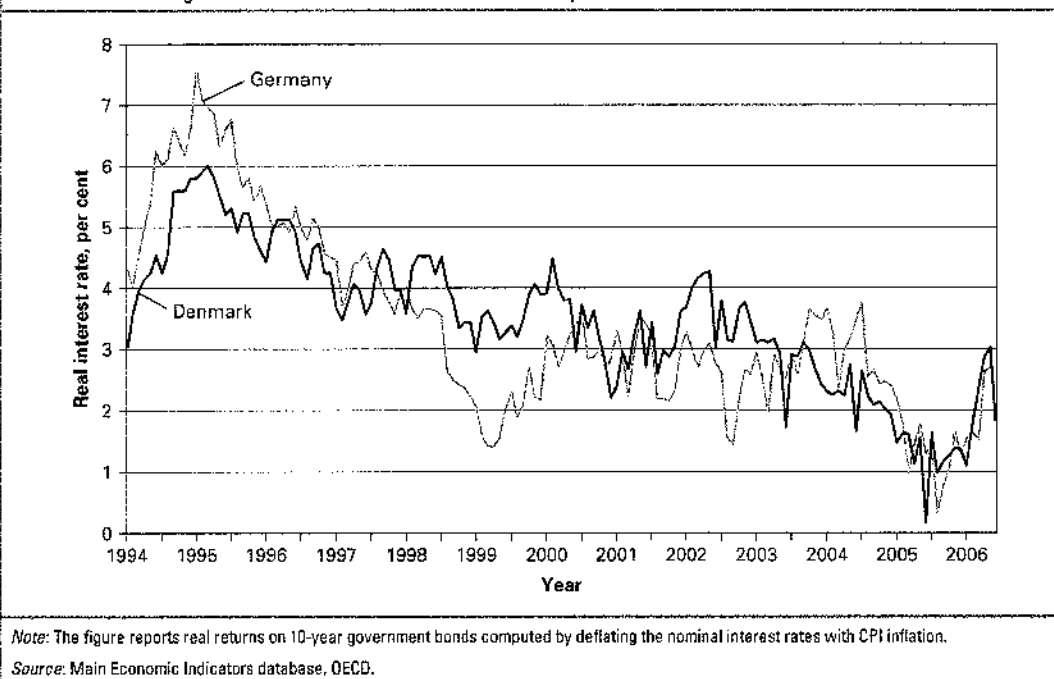
The open economy considered is so small that its own economic activity does not significantly affect economic conditions in the rest of the world. In particular, the economy is so small that the volume of domestic saving and investment does not have any noticeable impact on world interest rates. Capital is assumed to be *perfectly mobile* between the domestic economy and the rest of the world. Under perfect capital mobility domestic and foreign assets are perfect substitutes, and domestic as well as foreign investors can instantaneously and costlessly switch between domestic and foreign assets. For concreteness, let us assume that foreign as well as domestic firms finance their investment in real capital by issuing bonds in the international capital market (alternatively we might assume that part of business investment is financed by issuing shares which are perfect substitutes for bonds in the eyes of financial investors). Under perfect capital mobility all of these bonds must then pay the same rate of interest. For example, if domestic bonds had a lower return than foreign bonds, international financial investors would immediately sell domestic bonds in order to buy foreign bonds, thereby driving down domestic bond prices until the rates of return were equalized. Given that this arbitrage can occur instantaneously and costlessly, bond returns (and hence effective foreign and domestic interest yields) must be equalized at any point in time.

For the small open economy this means that the domestic real interest rate is effectively given by the world real interest rate which the domestic economy cannot affect. Denmark is a good example of a small open economy in which the interest rate is more or less given from abroad. As shown in Fig. 4.3, the real rate of interest on Danish long-term bonds has followed the German real interest rate quite closely in recent years.³

Just as foreign and domestic assets are taken to be perfect substitutes, we assume that the goods produced in the small domestic economy are perfect substitutes for (some of the) goods produced in other countries. With free trade in commodities this means that the prices of domestic goods must equal the prices determined in the world market (with the possible addition of some exogenous transport cost). From the viewpoint of the small domestic economy the prices of exported as well as imported goods and services are thus exogenously given from abroad, that is, the ratio of the domestic to the foreign price level is exogenously fixed. For simplicity we will therefore set this exogenous relative price equal to unity. This is equivalent to assuming that all economies in the world produce the same single good.

3. Denmark has maintained fixed exchange rates vis-à-vis the German D-mark, and subsequently against the euro. As we shall see in Part 7, when exchange rates are flexible the domestic *nominal* interest rate may deviate from the world interest rate because exchange rate changes (currency gains or losses) are part of the total return on foreign bonds. However, with flexible exchange rates there will still be a tendency for the domestic *real* interest rate to follow the foreign *real* interest rate, at least over the longer run (as will be explained in detail in Part 7).

FIGURE 4.3 Real long-term interest rates in Denmark and Germany



With this assumption our model cannot explain the international pattern of commodity trade (i.e., which countries end up producing which types of goods) and the welfare gains from such trade. Our intention here is to understand the implications of international capital mobility for wealth accumulation and the potential gains from capital mobility.

Setting up the model

We are now ready to specify our long-run model of the small open economy. Since the rate of depreciation of the capital stock, δ , will not play any important role in our analysis, we will set it equal to zero. The intertemporal budget constraint, (1), then simplifies to:

$$F_{t+1} - F_t + K_{t+1} - K_t = S_t$$

In the open economy it is important to distinguish between the country's *capital stock* and its total national *wealth*. National wealth, V_t , at the beginning of period t is the sum of the capital stock and the country's stock of net foreign assets at that time:

$$V_t \equiv K_t + F_t \quad (7)$$

The capital stock is wealth invested at home by domestic citizens or foreigners, while net foreign assets are wealth invested abroad by domestic citizens minus the part of domestic capital owned by foreigners. With this definition, the intertemporal budget constraint may be written as:

$$V_{t+1} - V_t = S_t \quad (8)$$

stating the simple fact that current savings imply a corresponding addition to national wealth.

As we have already seen, with capital mobility it is also necessary to distinguish between GDP, denoted by Y_t , and national income or GNP, denoted by Y_t^p . National income is defined as:

$$Y_t^n \equiv Y_t + \bar{r}F_t, \quad (9)$$

where $\bar{r}$ is the world real interest rate which is also the domestic real interest rate, given our assumption of perfect capital mobility. Since we are treating $\bar{r}$ as an exogenous constant, this variable does not carry any time subscript.

In accordance with the basic Solow model, we also assume a constant savings rate, s , but now aggregate national saving is a constant fraction of aggregate national *income* rather than a fraction of domestic product, since income from abroad can also be saved:

$$S_t = sY_t^n, \quad 0 < s < 1. \quad (10)$$

Following the basic Solow model, aggregate output is given by the Cobb–Douglas production function:

$$Y_t = BK_t^\alpha L_t^{1-\alpha}, \quad B > 0, \quad 0 < \alpha < 1, \quad (11)$$

and the labour force, L_t , is growing at the constant rate n :

$$L_{t+1} = (1+n)L_t, \quad n > -1. \quad (12)$$

Maximizing profits under perfect competition, the representative domestic firm employs capital and labour up to the point where marginal products are equal to the real factor prices. Since we abstract from depreciation, the real user cost of capital (the rental rate) is simply equal to the real interest rate, $\bar{r}$. Hence we have:

$$\bar{r} = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1}, \quad (13)$$

$$w_t = (1-\alpha)B \left(\frac{K_t}{L_t} \right)^\alpha. \quad (14)$$

THE COMPLETE SOLOW MODEL OF THE SMALL OPEN ECONOMY



$$V_t \equiv K_t + F_t, \quad (7)$$

$$V_{t+1} - V_t = S_t, \quad V_0 \text{ given} \quad (8)$$

$$Y_t^n \equiv Y_t + \bar{r}F_t, \quad (9)$$

$$S_t = sY_t^n, \quad (10)$$

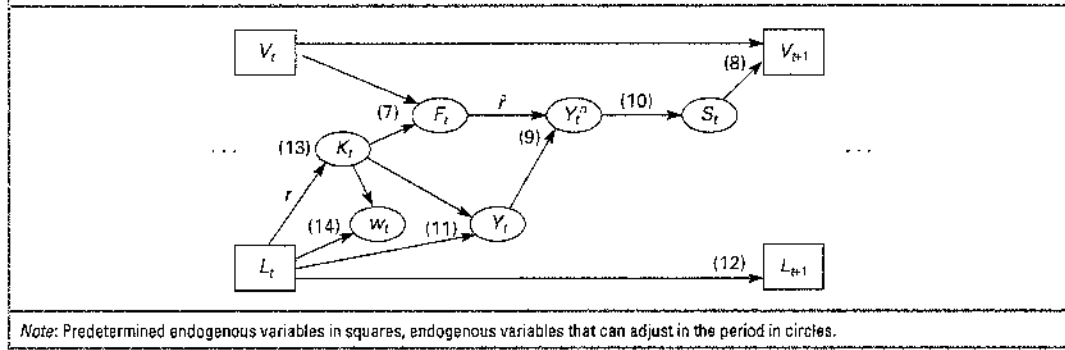
$$Y_t = BK_t^\alpha L_t^{1-\alpha}, \quad (11)$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given} \quad (12)$$

$$\bar{r} = \alpha B \left(\frac{K_t}{L_t} \right)^{\alpha-1}, \quad (13)$$

$$w_t = (1-\alpha)B \left(\frac{K_t}{L_t} \right)^\alpha. \quad (14)$$

Parameters: α , B , s , n and $\bar{r}$. Endogenous variables: V_t , L_t , K_t , F_t , Y_t , w_t , Y_t^n and S_t .

FIGURE 4.4 The dynamics of the Solow model for the small open economy

The dynamics of the model are illustrated in Fig. 4.4. The predetermined variables in period t are national wealth, V_t , and the labour force, L_t , given from past accumulation of savings and from past population growth, respectively. The capital stock, K_t , is *not* predetermined since it can adjust within period t by capital export or import. Equation (7) shows that a larger capital stock, K_t , is associated with a lower value of net foreign assets, F_t . Given L_t , the domestic capital stock adjusts to the level K_t given by (13), ensuring that the marginal product of capital equals the world real interest rate, $\bar{r}$. Suppose for a moment that the domestic capital stock were lower than this level, implying a marginal product of capital above $\bar{r}$. The return to capital invested in the domestic economy would then be higher than the return offered in the international capital market. Capital would therefore flow into the domestic country, creating an increase in the capital stock until the marginal product of capital just became equal to $\bar{r}$. In this way (13) determines the domestic capital–labour ratio, K_t/L_t . With K_t determined and with the predetermined L_t , domestic output, Y_t , is determined by (11), and the marginal product of labour, and hence the real wage, is determined by (14). Given V_t and K_t , the net foreign asset position, F_t , follows from (7). With Y_t and F_t determined, national income, Y_t^n , is determined by (9), and Y_t^n , in turn, determines national savings, S_t , through (10). National savings add to national wealth determining V_{t+1} through (8), while L_{t+1} follows from L_t by (12). We can then start over with period $t + 1$, etc. From given initial values, V_0 and L_0 , our model thus determines the dynamic evolution of all the endogenous variables.

4.3 Analysing the model: dynamics and steady state

As in Chapter 3 we will analyse our model in terms of per capita (or worker) variables. Using the notation:

$$y_t \equiv \frac{Y_t}{L_t}, \quad k_t \equiv \frac{K_t}{L_t}, \quad y_t^n \equiv \frac{Y_t^n}{L_t}, \quad v_t \equiv \frac{V_t}{L_t}, \quad f_t \equiv \frac{F_t}{L_t}. \quad (15)$$

Here, y_t is domestic product per capita, while y_t^n is national income per capita. Likewise, k_t is domestic capital per capita, while v_t is national wealth per capita, and f_t is net foreign assets per capita. It follows from (7) that $f_t = v_t - k_t$, and from (9) that $y_t^n = y_t + \bar{r}f_t$. The production function, (11), implies $y_t = Bk_t^\alpha$.

National income per capita

Equation (13) can be written $\bar{r} = \alpha Bk_t^{\alpha-1}$, and hence $\bar{r}k_t = \alpha Bk_t^\alpha = \alpha y_t$. Furthermore, from (14), $w_t = (1 - \alpha)Bk_t^\alpha = (1 - \alpha)y_t$. It follows that $y_t = \bar{r}k_t + w_t$. Thus GDP is fully used to reward the

labour and the capital employed in the country, and the share of labour income in GDP, w_t/y_t , is $1 - \alpha$, while the income share of domestic capital in GDP, $\bar{r}k_t/y_t$, is α . These features are unchanged from the model for the closed economy of Chapter 3, but note that the share of labour income in *national* income, w_t/y_t^n , is not (necessarily) $1 - \alpha$. An exercise will ask you to analyse the income distribution further.

National income per capita is $y_t^n = y_t + \bar{r}f_t = y_t + \bar{r}(v_t - k_t) = y_t - \bar{r}k_t + \bar{r}v_t$. Using here that $y_t - \bar{r}k_t = w_t$ gives:

$$y_t^n = w_t + \bar{r}v_t. \quad (16)$$

In the open economy, national income (per capita) is the sum of wages (per capita) and the return to national wealth (per capita). This reflects that the labour force and the capital *owned* by the citizens in the country (not the capital *employed* in the country) are the country's sources of income.

The law of motion

We described above how capital exports or imports ensure that the capital stock and capital per worker adjust *within each period* to ensure that the marginal product of capital equals the international interest rate, $\bar{r} = \alpha B k_t^{\alpha-1}$. In other words, k_t adjusts *immediately* to the particular constant level k^* given by $\bar{r} = \alpha B (k^*)^{\alpha-1}$. The constant (steady state) value for the capital intensity is thus:

$$k^* = B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{1}{1-\alpha}}. \quad (17)$$

Consequently, domestic output per worker, $y_t = B k_t^\alpha$, and the wage rate, $w_t = (1 - \alpha) B k_t^\alpha$, must jump immediately to the steady state levels:

$$y^* = B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}, \quad (18)$$

$$w^* = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}, \quad (19)$$

and the capital–output ratio must jump to:

$$\left(\frac{K_t}{Y_t} \right)^* = \frac{k^*}{y^*} = \frac{\alpha}{\bar{r}}. \quad (20)$$

The constancy of the domestic capital intensity, k_t , and of GDP per capita, y_t , means that the capital stock, K_t , and the domestic product, Y_t , both grow steadily over time at a rate equal to the rate of population growth, and this occurs immediately without any adjustment time required (for the closed economy similar conditions held in the steady state, but not during the adjustment to steady state). In particular, $K_{t+1} = (1 + n)K_t$, so $I_t = K_{t+1} - K_t = nK_t$. It follows that the rate of accumulation of domestic capital, I_t/K_t , adjusts immediately to the constant level n , and, using (20), that the investment rate, $I_t/Y_t = n(K_t/Y_t)$, adjusts immediately to:

$$\left(\frac{I_t}{Y_t} \right)^* \equiv i^* = \frac{n\alpha}{\bar{r}}. \quad (21)$$

Thus, if cross-country variations in the ratio S_t/Y_t are driven mainly by variations in s , our Solow model of the small open economy suggests that the parameter α_1 in the Feldstein–Horioka equation (3) should be close to zero (since $(I_t/Y_t)^*$ is independent of s).

These properties contrast with the basic Solow model for the closed economy where capital and output per worker as well as factor prices depend on the domestic savings rate, s (in steady state and during the adjustment to steady state), and where capital and output per worker, and hence the rate of capital accumulation, only gradually converge on their steady state levels.

Instead of governing *capital* accumulation, the propensity to save now influences the accumulation of national *wealth*. To see this, start from Eq. (8), $V_{t+1} = V_t + S_t$, divide through by $L_{t+1} = (1+n)L_t$, and use (9) and (10) to obtain:

$$v_{t+1} = \frac{1}{1+n}(v_t + sy_t^n).$$

Now insert the expression in (16) for y_t^n , and use $w_t = w^*$ to get

$$v_{t+1} = \frac{1}{1+n}(v_t + s(w^* + \bar{r}v_t)),$$

from which we obtain:

$$v_{t+1} = \frac{1+s\bar{r}}{1+n}v_t + \frac{sw^*}{1+n}. \quad (22)$$

Equation (22) is the *transition equation for the small open economy*. We have chosen to express the transition equation in terms of w^* , which is constant as long as $\bar{r}$ remains constant, but one can trace the transition equation back to parameters by substituting (19) for w^* in (22).

Equation (22) is quite intuitive. The sources of national wealth per capita in period $t+1$ are the national wealth carried over from period t , adjusted for population growth, $v_t/(1+n)$, plus the savings per capita in period t , $sy_t^n = s(w^* + \bar{r}v_t)$, adjusted for population growth, $s(w^* + \bar{r}v_t)/(1+n)$.

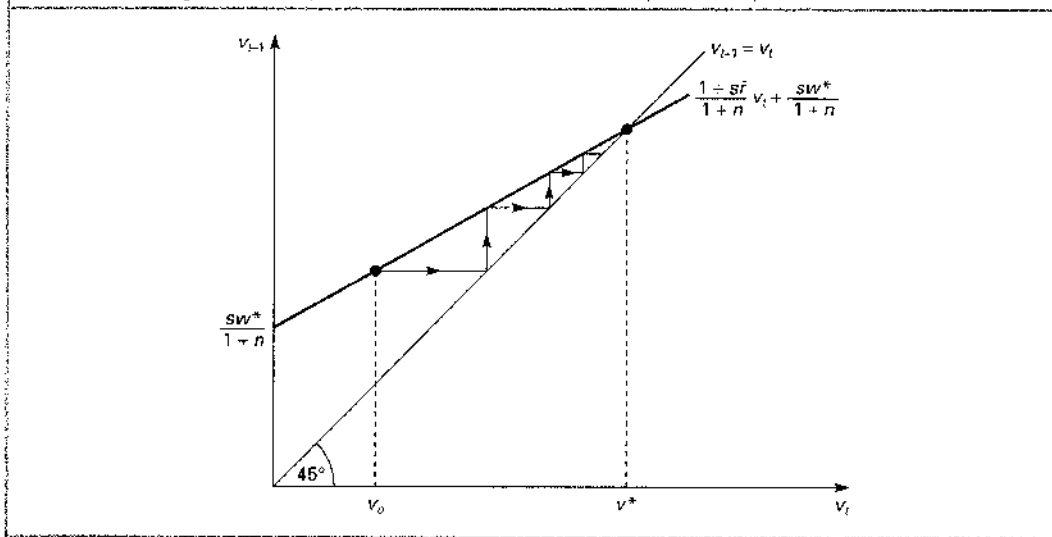
Given an initial value, v_0 , of wealth per capita, the transition equation determines v_1 , which in turn determines v_2 , etc., thus determining the full sequence (v_t) . From (v_t) , the sequences for all the endogenous variables may then be derived. For instance, the sequence for national income, (y_t^n) , follows from $y_t^n = w_t + \bar{r}v_t$, (16) above, by using $w_t = w^*$ for all t . Then the sequence, (c_t) , for consumption per capita follows from $c_t = (1-s)y_t^n$.

Convergence to steady state

The transition equation is illustrated in the transition diagram, Fig. 4.5. The transition curve is a straight line that intersects the vertical axis at $sw^*/(1+n) > 0$, and has a slope of $(1+s\bar{r})/(1+n)$. If this slope is smaller than one, as drawn, there is a unique positive intersection with the 45° line at a value for v_t denoted by v^* . From any predetermined initial wealth, $v_0 > 0$, v_t will converge towards v^* , as indicated by the iterations illustrated by the ‘staircase’ in Fig. 4.5. The convergence point, v^* , is the steady state value for national wealth per capita.

For the closed economy considered in Chapter 3, the transition curve (describing the accumulation of capital) was concave, reflecting diminishing returns to capital. In the small open economy, independently of how much wealth has already been accumulated, additional wealth always earns the international rate of return, $\bar{r}$. Hence, although there are diminishing returns to domestic capital, an increase Δv in current national wealth always creates the same increase $\bar{r}\Delta v$ in current national income and the same increase $s\bar{r}\Delta v$ in current savings, thus creating the same addition $(1+s\bar{r})\Delta v/(1+n)$ to national wealth next period after

FIGURE 4.5 Convergence to steady state in the Solow model of the small open economy



adjusting for population growth. This is why the transition curve is a straight line with slope $(1 + s\bar{f})/(1 + n)$.

Figure 4.5 shows that the economy will converge on the steady state *only* if the transition line is flatter than the 45° line,⁴ that is, if $(1 + s\bar{f})/(1 + n) < 1$. This requires fulfilment of the *stability condition*:

$$s\bar{f} < n. \quad (23)$$

In Western countries a long-run national savings rate around 0.2 is fairly common, and over the long run real interest rates in most Western economies have tended to fluctuate around a level of 3–4 per cent per annum, so a value of $s\bar{f}$ around 0.007 would not be unrealistic. In a literal interpretation of our model, n is the rate of population (labour force) growth which is rarely as high as 0.7 per cent per year in OECD countries (see Table A at the end of Book One). This seems to suggest that the stability condition (23) may not be met in practice.

However, this is an artefact of disregarding technological progress. In a more realistic setting with technological progress, we should measure the labour force, L_t , in so-called ‘efficiency units’, accounting for the fact that the rising level of technology gradually makes each unit of labour more productive. Effective labour input would then grow at a rate larger than the growth rate of the population and the stability condition would be just like (23) with n interpreted as the growth rate of effective labour, which is the sum of the rate of population growth and the rate of so-called ‘labour augmenting’ technological progress. Chapter 5 will present this extension of the Solow model in detail and explain, among other things, that the economy’s long-run growth rate in GDP equals the growth rate of effective labour supply. Since long-run GDP growth rates in Western economies are considerably above 0.7 per cent per year, the stability condition would be met.

An appropriate way of evaluating (23) in the present context without technological progress is this: assume for a moment that the small economy considered is completely closed to the outside world. It should then work according to the model of Chapter 3, converging to a steady state in which the key variables would be given by the formulae in Section 3.3. In particular, the real interest rate established in the long run would be (recall that $\delta = 0$)

4. You may check this by doing a graphical analysis of the case where the transition line is steeper than the 45° line.

$$r_c^* = \frac{\alpha n}{s}, \quad (24)$$

where the subscript c refers to 'closed'. The world economy itself is closed, so in the long run the international interest rate should be $\bar{r} = \alpha \bar{n} / \bar{s}$, where $\bar{n}$ and $\bar{s}$ are the average international population growth rate and the international savings rate, respectively (and we have assumed a common α). If the small economy considered is structurally identical to the world economy ($n/s = \bar{n}/\bar{s}$), then $r_c^* = \bar{r}$, and (23) becomes equivalent to $\alpha < 1$, which is indeed one of our assumptions (realistically, α is much smaller than one, around $\frac{1}{3}$). In other words, as long as just the small economy we consider is not very different from the surrounding world economy (structurally), the stability condition, $s\bar{r} < n$, will be fulfilled.

Without apology, we will therefore assume that (23) is indeed satisfied, implying that v_t converges to the steady state value v^* . Consequently, since $y_t^n = w_t + \bar{r}v_t$, national income per capita must converge to $y^{n*} = w^* + \bar{r}v^*$, and consumption per capita, c_t , converges to the steady state level, $c^* = (1 - s)y^{n*}$.

The key endogenous variables in steady state

Setting $v_t = v_{t+1} = v^*$ in (22) and solving for v^* , we find the steady state level of national wealth per capita,

$$v^* = \frac{s}{n - s\bar{r}} w^* = \frac{\frac{s}{\bar{n}}}{1 - \frac{s}{\bar{n}}\bar{r}} w^*, \quad (25)$$

which is positive under our stability condition, $(s/n)\bar{r} < 1$. The expression for w^* found in (19) can be inserted in place of the w^* in (25) to trace v^* back to parameters. Since w^* is independent of s and n , we see that the larger s/n is, the larger steady state national wealth per capita will be, reflecting that more savings and less erosion of per capita wealth by population growth both tend to create more wealth per person.

The net foreign asset position in steady state is $f^* = v^* - k^*$. From our solutions (17) and (19) for k^* and w^* , it follows that $k^* = (\alpha/\bar{r})w^*/(1 - \alpha)$. Inserting this expression for k^* along with the expression in (25) for v^* into $f^* = v^* - k^*$, we get:

$$f^* = \frac{1}{1 - \alpha} \frac{s}{n} \frac{1}{\bar{r}} \frac{\bar{r} - \frac{\alpha n}{s}}{1 - \frac{s}{\bar{n}}\bar{r}} w^*. \quad (26)$$

This equation reveals that under our stability condition, net foreign assets are positive in steady state exactly if $\alpha n/s < \bar{r}$, while f^* is negative if $\alpha n/s > \bar{r}$. From (24), the condition $\alpha n/s < \bar{r}$ translates into $r_c^* < \bar{r}$. Hence the small open economy will become a net creditor if and only if the long-run interest rate it would create if it were closed is lower than the world interest rate. This is quite intuitive. If a country starts out from autarky, an opening of the capital account should lead to a capital export, and hence to a positive net foreign asset position, if the international capital market offers a higher rate of return than the domestic economy prior to the liberalization of capital flows. Conversely, if a country offers a rate of return above the international level before capital mobility is allowed, an opening of the capital account should lead to a capital import. Consequently, a country with a relatively strong propensity to save ($s/n > \bar{s}/\bar{n}$ and thus $r_c^* < \bar{r}$) will end up as a net creditor in the international economy, while a country with a relatively weak propensity to save ($s/n < \bar{s}/\bar{n}$) will become a net debtor.

The values for variables such as domestic capital per capita, GDP per capita, the wage rate, etc., that jump immediately to their steady state levels have been stated in (17)–(21)

above. The steady state value for national income per capita is $y^{n*} = w^* + \bar{r}v^*$, which we rewrite by inserting the expression in (25) for v^* :

$$y^{n*} = w^* + \bar{r} \frac{s}{n - s\bar{r}} w^* = w^* \left(1 + \frac{s\bar{r}}{n - s\bar{r}} \right) = w^* \frac{n}{n - s\bar{r}} = w^* \frac{1}{1 - \frac{s}{n}\bar{r}}.$$

We can then trace all the way back to parameters by inserting our solution, (19), for w^* :

$$y^{n*} = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left(\frac{\alpha}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \cdot \frac{1}{1 - \frac{s}{n}\bar{r}}. \quad (27)$$

Note that y^{n*} is positive under our stability condition, $(s/n)\bar{r} < 1$. The terms in front of the dot on the right-hand side of (27) reflect the influence from the wage rate, w^* : the lower the world interest rate, $\bar{r}$, the more capital per worker will be built up in the domestic economy, and the higher the wage rate will be. The term after the dot in (27) includes the influence of the income from wealth, $\bar{r}v^*$. Here a low value of $\bar{r}$ has a negative influence, whereas a large value of the ratio s/n has a positive influence because it generates a high level of wealth per capita, v^* .

The steady state value for consumption per capita may be found as $c^* = (1 - s)y^{n*}$. In this model there is no meaningful golden rule value for the domestic savings rate, as you will be asked to explain in an exercise.

Issues of structural policy

Among the model parameters that domestic policy authorities should be able to influence are s and n . According to (27) a high value of s/n tends to create a high value of national income per capita. Hence, with respect to our fundamental question, ‘what can make a nation rich?’, our conclusions are qualitatively unchanged from Chapter 3. Attaining a high national savings rate and getting population growth under control will have a positive impact on income per capita in open as well as closed economies, reflecting that a high level of national wealth per capita will always contribute to a high level of national income. It is even true that the elasticity of steady state national income with respect to the savings rate is not too different between the closed economy and the small open economy as long as the latter is not too different structurally from the world economy. You will be asked to verify this in Exercise 1.

Likewise, an improvement of technology (an increase in B) increases income per capita in the open as well as in the closed economy.

CONVERGENCE, STEADY STATE AND STRUCTURAL POLICY



Under a plausible stability condition, the Solow model for the small open economy implies convergence to a steady state. Economies with relatively strong savings propensities become net creditors in steady state, while economies with relatively weak savings propensities become net debtors. The structural policies required to generate a high long-run income per capita are similar in both the small open and the closed economy, and are policies ensuring a high savings ratio, a low population growth rate and a high technological level.

There is, however, an important structural policy issue which is not just a matter of appropriate values for structural parameters: should an economy choose to open itself to the outside world or is it better to be closed?

The open economy exposes itself to one type of shock that a closed economy avoids, namely international interest rate shocks. Does this feature represent a serious risk? And will a country benefit in the long run from an open capital account by obtaining higher income per capita? The answers to such questions are obviously important for decisions to liberalize capital flows.

4.4 Implications of an open capital account: advantages and drawbacks of globalization

Capital mobility, national income and the income distribution: a basic theorem

Will a country get richer if its government allows free capital mobility between the domestic economy and the rest of the world? In the first decades after the Second World War almost all governments in the capitalist world maintained strict capital controls.⁵ As we have seen, capital was later on allowed to move much more freely between countries. Our first question is: from the viewpoint of the individual small open economy, should we expect the liberalization of capital movements to have contributed positively to national income per capita in the long run?

We will analyse this important policy issue by comparing the long-run level of national income under free capital mobility to the long-run income level attainable in a policy regime of 'autarky' where the government does not allow international capital flows. Our answer will be: yes, an open capital account does create a higher *average* income in a small economy, but domestic *wages* will fall in consequence of an opening if the domestic economy has a relatively high propensity to save. We will first give the intuition for our results and then prove them formally.

If the small economy were closed, the real interest rate established in the long run would be $r_c^* = \alpha n/s$, as stated in (24) above. Except for a knife-edge case, there are two possibilities. Either the interest rate that the small economy would experience if it were closed is lower than the international interest rate, or it is higher, that is, either $r_c^* < \bar{r}$ or $r_c^* > \bar{r}$. The first case materializes when the small economy has a relatively high savings rate relative to population growth, s/n , while the latter case corresponds to a relatively low s/n -value.

Consider the case $r_c^* < \bar{r}$, and assume that the small economy decides to open up and allow free movements of capital and commodities. The domestic owners of capital will then be able to earn more by investing their wealth in the international capital market rather than placing it in domestic capital as they had to do before the opening of the capital account. Hence, a country with a relatively high propensity to save will benefit from capital mobility by exporting capital, thus taking advantage of the higher international interest rate and earning more on its national wealth by becoming an international net creditor. After the international reallocation of portfolios has been completed there will no longer be a difference between the return on domestic capital and the international interest rate. As capital flows out of the domestic economy, domestic capital per worker decreases, creating a higher domestic marginal product of capital. This does not change the fact that domestic wealth owners benefit from the opening through a higher rate of return on wealth. However, as capital per worker decreases, the marginal product of labour goes down. Hence the capital account liberalization implies lower real wages in the domestic economy. If the small economy has a high savings propensity an opening should thus imply higher average income, but those who own no wealth and must live from their labour alone will be hurt.

Next consider the case $r_c^* > \bar{r}$. In this case an opening of the capital account will benefit the small economy via capital *import*, making it a net debtor. Since the rate of return on domestic capital is higher than the international interest rate, domestic residents can make

5. By decoupling the domestic interest rate from foreign interest rates, capital controls enable a government to pursue an independent national monetary policy even if exchange rates are fixed.

money by borrowing in the international capital market and investing the borrowed funds in domestic capital. This arbitrage is advantageous until the two rates of return have been equalized by the resulting increase in the domestic capital intensity which will imply higher domestic wages. In this case capital account liberalization will benefit domestic workers, but domestic wealth owners will be hurt by the fall in the rate of return on their wealth.

In the knife-edge case, $r_c^* = \bar{r}$, an opening of the capital account should have no influence on (steady state) average income or wages in the home country, according to our reasoning.

To prove formally that an open capital account always implies a higher long-run average income, and lower (higher) wages if the economy has a relatively high (low) savings propensity, we import expressions from Section 3.3 stating what national income per capita and the wage rate would be if the small economy were closed:

$$y_c^* = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n} \right)^{\frac{\alpha}{1-\alpha}}. \quad (28)$$

$$w_c^* = (1-\alpha) B^{\frac{1}{1-\alpha}} \left(\frac{s}{n} \right)^{\frac{\alpha}{1-\alpha}}. \quad (29)$$

Define x as national income per capita under an open capital account *relative to* national income per capita under a closed capital account, and use our expressions for y^{n*} (in (27)) and for y_c^* to get:

$$x \equiv \frac{y^{n*}}{y_c^*} = (1-\alpha) \left(\frac{1}{\bar{r}} \frac{\alpha n}{s} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - \frac{s}{n} \bar{r}}.$$

From (24) we can replace $\alpha n/s$ by r_c^* , and s/n by α/r_c^* giving:

$$x = (1-\alpha) \left(\frac{1}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} \frac{1}{1 - \alpha \bar{r}}, \quad \text{where } \bar{r} \equiv \frac{\bar{r}}{r_c^*}. \quad (30)$$

We establish three features of x as a function of $\bar{r}$. First, if $\bar{r} = 1$ (corresponding to $\bar{r} = r_c^*$), we have $x = 1$, equivalent to $y^{n*} = y_c^*$. This confirms the intuitive insight that in case of $r_c^* = \bar{r}$, an opening of the capital account should have no influence on income per capita in the long run. Second, the derivative of $\ln x$ with respect to $\bar{r}$,

$$\frac{d \ln x}{d \bar{r}} = -\frac{\alpha}{1-\alpha} \frac{1}{\bar{r}} + \frac{\alpha}{1-\alpha \bar{r}},$$

is zero for $\bar{r} = 1$. Third, the same derivative is strictly increasing in $\bar{r}$, since we assume the stability condition, $(s/n)\bar{r} < 1$, now translating into $\alpha \bar{r} < 1$.⁶ These three properties imply that x as a function of $\bar{r}$ attains its minimum for $\bar{r} = 1$, and the minimum value is one. In other words, for any constellation of r_c^* and $\bar{r}$ other than $r_c^* = \bar{r}$, y^{n*} is larger than y_c^* (under our realistic stability condition). An open capital account thus implies a higher average income in the long run, except in the knife-edge case, $r_c^* = \bar{r}$.

Next consider the wage rate under an open capital account relative to the wage rate under a closed capital account. From (19) and (29) you will easily find that

6. Recall from (24) that $r_c^* = \alpha n/s$, so $\bar{r} = \bar{r}/r_c^* = s\bar{r}/\alpha n$, implying $\alpha \bar{r} = s\bar{r}/n$. Hence the stability condition $s\bar{r}/n < 1$ is equivalent to $\alpha \bar{r} < 1$.

$$\frac{w^*}{w_c^*} = \left(\frac{\frac{\alpha n}{s}}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}} = \left(\frac{r_c^*}{\bar{r}} \right)^{\frac{\alpha}{1-\alpha}}, \quad (31)$$

from which follows that $w^* < w_c^*$ exactly if $r_c^* < \bar{r}$, which is the same as $\alpha n/s < \bar{r}$. The latter $\bar{r}$ could be replaced by $\alpha \bar{n}/\bar{s}$, so the condition would read, $n/s < \bar{n}/\bar{s}$ confirming that an opening of the capital account will lead to a fall (rise) in wages in a country with a relatively high (low) savings rate.

LONG-RUN IMPLICATIONS OF AN OPEN CAPITAL ACCOUNT



If a small economy opens its capital account, allowing capital mobility to eliminate the difference between the domestic and the international real interest rate, the long-run implications according to the Solow model are: (i) Income per capita will increase. (ii) If the economy has a relatively high savings capacity, capital will flow out of the economy making it a net creditor, and the domestic capital intensity will fall, implying a higher rate of return on domestic capital and lower domestic real wages. (iii) If the economy has a relatively low savings capacity, capital will flow into the economy, making it a net debtor, and the domestic capital intensity will increase, implying a lower rate of return on domestic capital and higher domestic real wages.

Discussing the basic theorem on capital mobility

How large are the potential long-run effects of liberalizing capital movements according to our model? As shown by (30) and (31) above, this depends on r_c^* , $\bar{r}$, and α . A reasonable calibration is $\bar{r} = 0.03$, and $\alpha = 1/3$. For a relatively poor closed economy with a low savings capacity, a plausible domestic real interest rate could be $r_c^* = 0.06$. You will easily compute from (30) and (31) that in these circumstances the long-run effects of opening up would be a 13 per cent increase of domestic average national income, and a 41 per cent increase of domestic real wages. These effects are already considerable. Assuming instead $r_c^* = 0.09$, which would not be far-fetched for a poor country, would give an increase in average income of 30 per cent and an increase in domestic real wages of 73 per cent. For a relatively rich economy we may assume $r_c^* = 0.02$, giving an 8 per cent increase in national income of opening up, but also an 18 per cent cut in real wages. These are again considerable effects, mainly for real wages.

A liberalization of capital flows should thus imply an increase in average income in the long run, and possibly of considerable size, but, as we have emphasized, different types of income will be affected in different directions. In particular, if the initial domestic interest rate is lower than the world interest rate, a liberalization of capital movements makes capital flow *out* of the country. Domestic capitalists will then gain from a higher return to wealth whereas workers will lose from the drop in real wages resulting from the fall in domestic capital intensity, and the loss may be considerable under plausible parameter values. If wealth is evenly distributed among the domestic residents, the drop in wages will be more than offset by the increase in income from wealth for each and every person. However, people living from labour income alone will experience a drop in income when a country with a high savings rate opens up its capital account. This may be a perfectly 'rational' reason for workers in relatively rich, high-saving countries to be concerned about the implications of globalization. Note, however, that since average income increases, it may be possible to design a policy of redistribution through taxes and transfers such that everybody gains in all circumstances.

Our basic theorem says that free capital movements should imply higher national income from the point of view of a single small economy. It follows by the same reasoning, however, that if the world consisted of a collection of small economies then, if every economy opened

its capital account, each and every economy would gain in the sense of attaining a higher national income. What is the basic intuition behind the feature that everybody can gain, that one country's gain is not another's loss? Where does the increase of the 'pie' to share basically come from? The answer is: from a more effective global allocation of capital. When all economies are closed, each will build up its own capital stock the size of which will reflect the economy's own savings capacity. Capital intensities will be different between different countries and therefore the marginal products of capital will be different in different countries. Whenever the marginal products are different in two different places, an increase in total production and income can be obtained by moving capital from where it has a low marginal product to where it has a high one without additional inputs being required. This is exactly what happens according to our theory when capital movements are liberalized, and the better use of the global capital stock explains why all countries can gain.

Just as international trade theory shows that free trade in goods and services generally tends to increase economic efficiency by enabling countries to exploit their comparative advantages, the theory of economic growth thus suggests that free trade in capital services (i.e. free international mobility of capital) tends to increase average long-run prosperity by implying a more effective global allocation of capital. This insight from our Solow models provides one explanation why most economists take a favourable attitude towards capital mobility, while realizing that redistributive policies may be needed to ensure that all groups in society enjoy the fruits of liberalization.

Our basic theorem is a pure steady state comparison. How fast do the effects of liberalization occur according to our theory? We have seen above that some variables adjust immediately, among these GDP per capita and real wages. So, the very important wage response should come fast. Some other variables have a gradual adjustment, among these national wealth and income per capita. An exercise will ask you to do simulations under plausible parameter restrictions for a relatively poor country to investigate how fast these more gradual effects occur. We may reveal that most of the adjustment of national income comes immediately and depends only a little on the gradual accumulation of national wealth. So, not only are the effects for a relatively poor country of opening its capital account beneficial and of considerable size (in particular for wages), they should also come very fast. We have to point to two reservations, however.

The very fast occurrence of the effects rests on our ideal assumption of *perfect* capital mobility. In the real world physical capital adjusts more slowly than required for an immediate equalization of marginal products across countries. In consequence the potential effects from liberalization come more slowly than our idealized model predicts.

Furthermore, when individual country risks are taken into account as we do below, the size and for some variables even the sign of the predicted effects of free capital mobility can be questioned.

Despite these reservations, our basic theorem does suggest a potential for highly beneficial effects of liberalizing capital movements. Moreover, in a more realistic setting with risk and uncertainty, capital mobility may even yield an additional type of welfare gain because it allows risk-averse savers to reduce the volatility (riskiness) of the rate of return on their wealth by holding foreign as well as domestic assets in their portfolios (assuming that the rates of return on foreign and domestic assets are not perfectly correlated).

On the other hand, a country opening its capital account exposes itself to a type of impact from the outside world that it would avoid if it remained closed, namely changes in the international interest rate, so-called interest rate shocks.

A permanent interest rate shock

It is of interest to analyse the *long-run* economic consequences of a *permanent* increase (or decrease) in the international interest rate, $\bar{r}$. Such an increase could, for instance, be caused by a permanent increase in government consumption in some of the larger countries in the

world. The world economic system is closed and, as we explained in Chapter 3, a higher government propensity to consume implies a higher long-run interest rate in a closed economy. Hence a permanent expansion of public consumption in the rest of the world will affect the small domestic economy via a permanent increase in $\bar{r}$. What are the implications according to our model?

Our formulae (17), (18) and (19) reveal directly that steady state domestic capital intensity, GDP per capita and the real wage all decrease in response to a permanent increase in $\bar{r}$. Just after the interest rate shock (before any adjustments have taken place), the marginal product of domestic capital, and hence the return earned on domestic capital, will fall short of the new and higher international interest rate. Hence capital flows out of the country, implying a decrease in domestic capital per worker. The outflow ceases when the decrease in k_t has driven the marginal product of capital up to the level of the new and higher $\bar{r}$. This adjustment of k_t takes place immediately and is effected by capital export. A capital export should imply that the country's net foreign asset position improves. We will return to this issue below.

The fall in domestic capital intensity explains why domestic output per capita falls. Furthermore, the lower value of k^* implies a smaller marginal product of labour and therefore a lower wage rate. The rate of return on domestic capital increases with $\bar{r}$, but note that capital income created in the domestic economy ($\bar{r}k^*$) actually decreases, as can be seen by multiplying both sides of (17) by $\bar{r}$.

How about national income? Consider the formula (27) for the long-run value of income per capita, and remember that the term in front of the dot on the right-hand side is the steady state wage rate, w^* . We see that this term decreases in response to the increase in $\bar{r}$, reflecting the fall in the marginal product of labour caused by the drop in domestic capital intensity. The term to the right of the dot includes the return on national wealth, $\bar{r}v^*$, and this term increases (under our stability requirement). There are thus two opposing forces on y^{n*} . To determine which one is the strongest, we can differentiate (the log of) y^{n*} with respect to $\bar{r}$:

$$\frac{d \ln y^{n*}}{d\bar{r}} = -\frac{\alpha}{1-\alpha} \frac{1}{\bar{r}} + \frac{\frac{s}{n}}{1-\frac{s}{n}}.$$

This expression includes a negative and a positive term, reflecting the two opposing forces. Under the stability condition, the condition for the derivative being negative is that $\bar{r} < an/s$. This is exactly the condition for the country being initially a net debtor, $\bar{r} < r_t^*$. On the other hand, the derivative is positive exactly if $\bar{r} > an/s$, that is, if the country is initially a net creditor. The conclusion is that in the long run national income (and consumption) per capita will fall in response to an increase in the world interest rate if the country has a low propensity to save and therefore is a net debtor, while national income per capita will increase if the country has a strong savings capacity and is therefore a net creditor.

This is easy to understand intuitively. If a country has debt, its average income will be negatively influenced if a higher rate of interest must be paid on the debt (and we would expect the debt to decrease after all adjustments have taken place). On the other hand, if a country has positive net foreign assets, the domestic wealth owners will benefit from a higher rate of return on wealth (and we would expect the country to react by increasing further its net foreign assets).

As already explained, domestic wage earners will experience a drop in their wage income regardless of the country's net foreign asset position and regardless of their individual asset positions: a person without any debt or wealth will lose income after an increase in the international interest rate as a consequence of the domestic economy's reactions to the interest rate shock.

How are the long-run values for national wealth per capita, v^* , and net foreign assets per capita, f^* , affected by the negative interest rate shock? In the derivation of (27) we found that

$$y^{n*} = \frac{w^*}{1 - \frac{s}{n}\bar{r}},$$

(last equation before (27)), so (25) may be written as

$$v^* = \frac{s}{n} y^{n*}.$$

This shows that v^* is affected by $\bar{r}$ through the same terms as y^{n*} , so in the long run national wealth falls after the increase in $\bar{r}$ if the economy has a relatively low propensity to save, and vice versa. National wealth is affected in the same direction as national income because wealth comes from income through savings.

Finally, we can use (24) and (25) to rewrite (26) as

$$f^* = \left(\frac{1}{1 - \alpha} \right) \left(\frac{\bar{r} - r_c^*}{\bar{r}} \right) v^*.$$

Thus f^* is influenced by $\bar{r}$ through the same terms as y^{n*} and v^* , and through the term $(\bar{r} - r_c^*)/\bar{r}$. If the country is initially a net debtor ($\bar{r} < r_c^*$), then this last term is negative, and as $\bar{r}$ increases it becomes smaller in absolute value. This reveals that net foreign assets per capita, f^* , altogether become smaller in absolute value. The domestic economy reacts to the increased price of debt by reducing its debt. On the other hand, if the country is initially a net creditor, an increase in $\bar{r}$ will make its positive net foreign assets even larger (show this).

In conclusion, a permanent interest rate shock where $\bar{r}$ increases has a negative influence on the domestic economy's long-run national income and wealth if the country is a net debtor and a positive influence if the country is a net creditor, while wages are negatively affected in all circumstances. A permanent decrease in $\bar{r}$ has, of course, just the opposite effects. If a country wants to avoid the ups and downs associated with international interest rate shocks, this may be a reason for not wanting to liberalize capital movements, although such a liberalization should have a positive long-run influence on national income on the average.

Country risk

So far we have assumed that the 'hurdle' rate of return required and accepted by investors for investing their wealth in the small economy is the same as the international interest rate, $\bar{r}$. However, in the real world investors will often require a *risk premium* on top of the international interest rate for investing in the assets of any particular small economy.

In August 1998 the American 'hedge fund', Long Term Capital Management, was in trouble (the reason is not important here) and had to sell off assets. It turned out that it had quite a lot of Danish mortgage bonds in its portfolio. Although the hedge fund was a small economic agent relative to the American or the European economy, it was relatively large compared to the Danish bond market and therefore its selling off caused a drop in Danish bond prices. This was bad news for the other investors who had (planned) to sell Danish bonds at that time. The wave could, of course, have gone the other way with investors being fortunate to sell Danish bonds at a time where large-scale buying by a big international investor had driven prices upwards. However, usually investors are risk averse, preferring a safe return of one million euros to a fifty-fifty chance of earning either two million euros or zero euros. Since the return on Danish assets can fluctuate a little depending on the particular circumstances of large international investors, investors in Danish bonds will require an expected return (slightly) in excess of the interest rate on the safest bonds traded in the international capital market. The 'thinness' of Danish asset markets may thus imply a

risk premium in the Danish interest rate. Figure 4.3 indicates that the Danish real interest rate has indeed been a bit higher than the euro interest rate over longer periods, but not much in recent years, though.

There can be other (more severe) reasons for risk premia in interest rates. For instance, if investment in a country is not considered completely safe because of risks of, for example, civil war, coup d'états, nationalizations or widespread bankruptcies, investors will typically require large risk premia for investing in the country.

The risk premium on a country's real interest rate can change, and the change can be long lasting, for instance if international investors lose confidence in a country. A dramatic example of this was provided by the South-East Asian economic crisis in 1997–98 when foreign capital flows into the 'tiger' economies of Indonesia, Malaysia, South Korea and Thailand suddenly stopped, forcing these countries into drastic cuts in domestic consumption and investment as they could no longer finance their large current account deficits.⁷ After such a loss of confidence investors sharply increase the risk premium they require in order to be willing to lend to the country.

In the great financial crisis of 2008 investors suddenly lost confidence in a lot of assets and financial institutions and capital therefore flowed massively into what was perceived to be risk-free assets issued by 'safe havens'. For this reason interest rates on the short-term government bonds of Germany and the USA fell to levels close to zero. At the same time the risk premia included in the interest rates of many other countries increased dramatically, not so much or not only because of reduced trust in these particular economies, but rather as a result of an extreme aversion to risk of any kind. Many so-called emerging market economies in Eastern Europe, South America and Asia experienced that capital fled their countries and that the real interest rates required to keep capital inside the country increased significantly.

Below we will analyse the effects of a long-lasting change in the risk premium on a country's real interest rate. Denoting, as before, the average international real rate of interest by $\bar{r}$, the hurdle real interest rate that investors require and accept for investing in a particular small economy may be $\bar{r} + \varepsilon$, where ε is a risk premium compensating for the risk associated with investing in the country's assets. In principle, ε could be negative if the country is considered very 'safe', but we will think of ε as positive.⁸

The risk premium, ε , can be influenced by certain types of domestic policy. For instance, if Denmark were to join the European Monetary Union, abandoning the Danish krone, Danish bonds denominated in euros would be very close substitutes for other euro bonds, and the market for Danish bonds would therefore no longer be thin. Hence, the risk we described above would be reduced and the (already small) risk premium in the Danish real interest rate should therefore also be reduced. Likewise, policies that promote the general safety of a country's assets, for instance an improvement of the financial system or of the system of corporate governance that reduces the risk of bankruptcies or fraud, should reduce the risk premium as well. Of course, a certain (bad) policy could imply an increase in the risk premium.

In our Solow model of the small open economy, (7)–(14), the interest rate, $\bar{r}$, appears in two places. One is in the arbitrage condition, (13), where $\bar{r}$ is obviously a hurdle interest rate and should be replaced by $\bar{r} + \varepsilon$ to take country risk into account explicitly. The other place is in the definition, (9), of national income. Should the $\bar{r}$ appearing here also be replaced by $\bar{r} + \varepsilon$?

7. In his highly readable book on *The Return of Depression Economics* (Allen Lane, The Penguin Press, 1999), Paul Krugman gives a breathtaking account of the Asian crisis as well as other recent international financial crises.

8. Just adding ε to $\bar{r}$ in the definition of the hurdle interest rate is an *ad hoc* way of dealing with country risks since it does not model uncertainty explicitly. Under uncertainty investors expect different rates of return from investment in the country with different probabilities. Since they are risk averse, they will require an *average* rate of return above the safe $\bar{r}$. We try to capture this by our ε , but without considering probability distributions explicitly.

This depends on the country's net foreign asset position. If it is a net debtor, $F_t < 0$, the foreign investors who have financed part of the country's capital stock will require a risk premium on their investment, so the interest rate that must be paid on F_t is indeed $\bar{r} + \varepsilon$. Hence, if the country is a net debtor, our model, (7)–(14), is amended to take country risk into account by writing $\bar{r} + \varepsilon$ instead of $\bar{r}$ wherever $\bar{r}$ appears.

If the country is a net creditor, it will earn only the international interest rate, $\bar{r}$ (without any risk premium), on its positive net foreign assets. In this case the appropriate model will therefore consist of (7)–(14), with the $\bar{r}$ in (9) kept unchanged, and with the $\bar{r}$ in (13) replaced by $\bar{r} + \varepsilon$.

We will consider a net debtor country that is initially in steady state. Our model is then simply (7)–(14), with $\bar{r}$ replaced by $\bar{r} + \varepsilon$. Mathematically the model is thus unchanged, with $\bar{r} + \varepsilon$ now taking the place of $\bar{r}$. We are interested in the long-run consequences of a permanent increase in the risk premium, ε . This situation could, for instance, be representative for a country that has developed by borrowing, thereby becoming a net debtor, and where some event suddenly causes a long-lasting lowering of investor expectations regarding the country's future long-term performance.

To analyse this situation we do not have to perform any new formal analysis. For a net debtor our model is unchanged with respect to mathematical structure, so in all our formulae for steady state values, transition curve, etc., we should just replace $\bar{r}$ by $\bar{r} + \varepsilon$. In the model with country risk, an increase in ε will have the same effects as an increase in $\bar{r}$ had in the old model, and for the same reasons. We have already analysed this and found that for a net debtor, wages, national income and national wealth would fall in the long run. Hence, this will be the long-run reaction of the small open debtor economy to an increase in the country risk premium, ε .

However, this time we cannot conclude that if the country is initially a net creditor, national income will increase in response to an increase in ε . Recall that the model including country risk is different for a net debtor and a net creditor. An exercise will ask you to analyse the case of a net creditor. The analysis will reveal that national income and wealth are negatively affected by an increase in the country risk premium in all circumstances. Summarizing with respect to interest rate and risk shocks:

LONG-RUN EFFECTS OF PERMANENT INTEREST RATE AND RISK PREMIUM SHOCKS



A higher international real interest rate will imply a higher national income in a small open creditor country and a lower income in a debtor country. Irrespective of the country's net foreign asset position, the domestic capital intensity and real wages will fall.

A higher risk premium on capital invested in a small open economy will cause the domestic real interest to increase and domestic national income, capital intensity and real wages to decrease.

The possibility of long-lasting interest rate shocks, caused by changes in international interest rates or by changes in country-specific risk premia, thus implies some risks for small open economies. However, the main risks of free capital movements lie in the economy's *short-run* reactions, and we should emphasize that our open economy Solow model cannot give a realistic description of the short-run effects of an interest rate shock. For example, although our model implies that workers must take a wage cut when the country is hit by an increase in $\bar{r}$ or ε , at least they keep their jobs. But in practice wages are not very flexible in the short run, so a higher interest rate will throw some people out of work for a while as firms reduce their investment.

As another example, in our model firms react to the higher interest rate by reducing their capital stock and output, but they 'stay in business'. However, in the real world a serious

interest rate shock will cause many firms to go bankrupt, and investment will fall more dramatically in the short run than in the long run.

In summary, because our Solow model with fully flexible prices does not allow output and employment to deviate from their 'natural' rates, it does not capture all the drama of the recession that will occur in the short run when a country runs into an international financial crisis. In Part 7 of this text we shall develop business cycle models which can illuminate the short-run fluctuations generated by such a shock.

4.5 The costs and benefits of capital mobility: second thoughts?

Much of the ongoing international debate on globalization centres on the question whether free capital mobility is desirable or not. The analysis in this chapter suggests that capital mobility brings long-run benefits as well as potential costs.

An open capital account should enable a low-saving country with high investment needs to borrow more cheaply in the world capital market. It also allows wealth owners in high-saving countries to take advantage of more profitable investment opportunities abroad. In these ways capital mobility tends to raise a country's long-run national income and wealth.

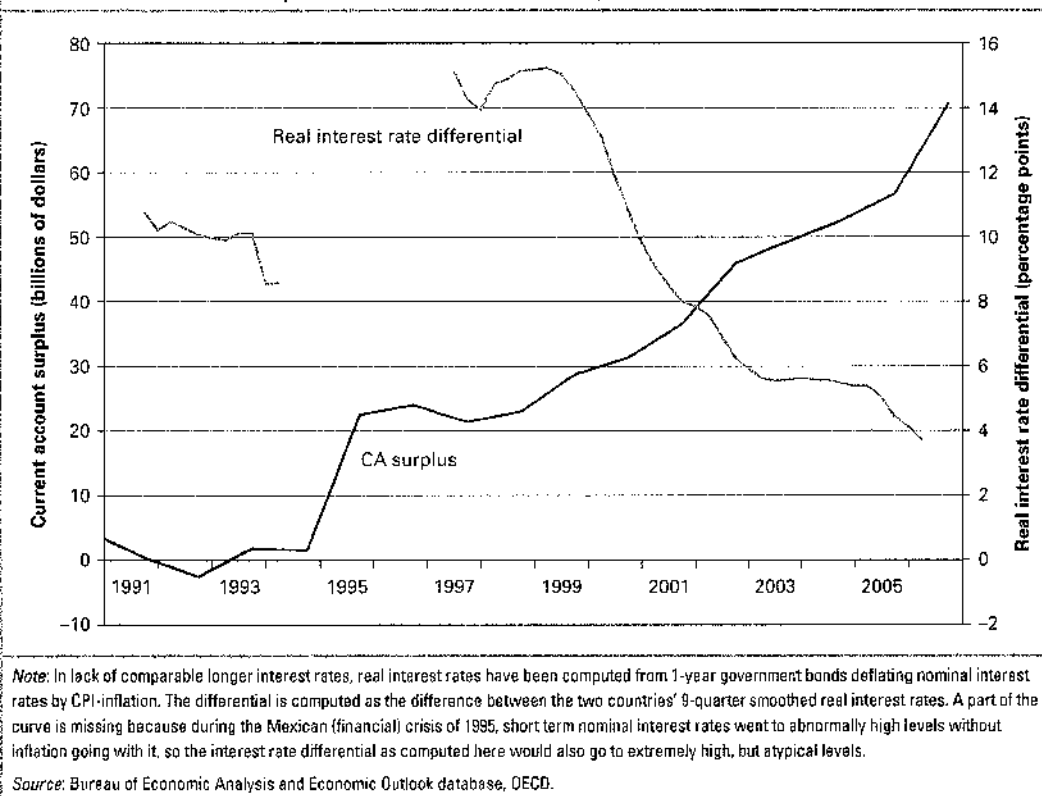
On the other hand capital mobility also makes a country vulnerable to shifts in the moods of international investors. If investors suddenly lose confidence in a debtor country, the resulting increase in the country's borrowing rate of interest will cause a recession and sometimes even a depression, as illustrated by the Asian crisis in 1997–98. Recent economic history suggests that developing countries and emerging market economies tend to experience greater shifts in international borrowing conditions than the most developed Western market economies. This may explain why many developing countries still maintain some capital controls whereas the richest countries in the world have liberalized international capital flows.

If the risks associated with *shifting* risk premia are not too serious, there should be a strong case for a small economy to open its capital account. In particular, for a low-saving, poor country the consequence should be an inflow of capital creating a higher domestic capital intensity and higher real wages to the benefit of the country's wage earners. Many economists have seen this wage effect as a fundamental argument in favour of capital mobility as a means for poor countries to alleviate poverty. However, many of these economists have had second thoughts, and the example of Mexico is a perfect illustration.

Consider Fig. 4.6. This shows that Mexico has typically had a positive real interest rate differential to the USA, and this is particularly true for the years up to 1994 (consult the figure's note about the difficulties of measuring the interest rate gap). This is in line with our theory since the USA has typically had an s/n -ratio more than twice that of Mexico (consult Table A in the appendix for Book One). Up to 1994 there was strict control on capital movements between the two countries, but in 1994 the North American Free Trade Agreement, NAFTA, between Canada, the USA and Mexico began to take effect. NAFTA involved liberalizations of trade as well as removal of restrictions on investment and capital movements between the countries. Economists expected that this would cause a huge flow of capital from the USA into Mexico, the flip side of which would be a current account *deficit* of Mexico vis-à-vis the USA. The figure shows that the exact opposite happened: Mexico more or less immediately began to run a huge *surplus* on the current account with the USA, so capital was flowing out of Mexico and into the USA. The relatively poor Mexican economy was thus helping the rich USA, already with abundant capital, to increase its capital intensity. The idea that free capital flows would help poor Mexican workers turned out not to hold.

Our model is compatible with this evidence. The outflow of capital from Mexico can be explained by country risk: Mexican as well as international investors placed such a large

FIGURE 4.6 Current account surplus and real interest rate differential, Mexico vs. USA



risk premium on investment in Mexico that, despite the apparently much higher Mexican real interest rates, investors considered investment in the USA more advantageous than investment in Mexico. When given the opportunity, rich Mexican wealth owners therefore began to move their capital to the USA. The perceived riskiness of investment in Mexico was confirmed by the Mexican crisis of 1995, but was probably also caused by a relative distrust in Mexican authorities and institutions at the time.

The Mexican case illustrates sharply that not only shifts in country risk, but also the presence of a large, constant country risk premium has important implications for the effects of capital mobility. In particular, one cannot take for granted that capital account liberalization will benefit the workers of poor labour-abundant countries through a capital inflow that raises the demand for labour.

Nevertheless, a modified version of our 'basic theorem' on capital mobility still applies even in the presence of country-specific risks. When Mexican wealth owners reacted to capital account liberalization by moving (some of) their capital to the USA, they did so because they expected a higher *risk-adjusted* return on investment in that country. Adjusted for risk, the net return to the capital owned by Mexicans therefore increased, and so did the 'risk-adjusted' Mexican national income, that is, national income adjusted for the welfare loss associated with investment risks. Even though Mexican GDP fell as capital fled the country, this was more than compensated by an increase in expected capital income from Mexican-owned investments in the USA and elsewhere, once the expected returns to capital were adjusted for risk. However, although total risk-adjusted income increased in Mexico as well as in the USA, the gains to Mexico were not equally shared, since Mexican workers faced less demand for their labour as capital was exported.

4.6 Summary



1. In an open economy which allows free international mobility of capital, part of domestic investment may be financed by capital imports from abroad, and part of domestic saving may be invested in foreign assets. It is then necessary to distinguish between the *domestic capital stock* (the stock of capital invested in the domestic economy) and *national wealth* (the total stock of assets owned by domestic residents). A country's stock of net foreign assets is equal to the difference between its national wealth and the domestic capital stock.
2. In parallel to the distinction between the domestic capital stock and national wealth, one must distinguish between *gross domestic product* (GDP) which is the total output produced by the factors of production employed in the domestic economy, and *national income* which is the total income earned by the factors of production owned by domestic residents. National income is equal to GDP plus the return to the country's net foreign assets.
3. The growth rate of a country's capital stock is given by its rate of domestic investment, whereas the growth rate of national wealth is determined by the national savings rate. In a closed economy saving and investment are identical and therefore perfectly correlated. In the open economy capital mobility reduces the correlation between saving and investment. The current account on the balance of payments represents the difference between national saving and national investment. As an accounting identity, the increase in a country's net foreign assets is given by the balance on its current account.
4. International capital mobility may take the form of international portfolio investment, where households or institutional investors (including banks) acquire foreign assets to diversify their portfolios, or it may take the form of foreign direct investment where the investor acquires a controlling ownership share of foreign business assets. In recent decades both forms of international investment have grown much more rapidly than world output and world trade, as many governments have liberalized capital flows, and as improvements in information and communication technologies have reduced the barriers to foreign investment. The growing mobility of capital has caused a fall in the correlation between national rates of saving and investment. In the euro area this correlation is now close to zero.
5. Capital is said to be perfectly mobile when domestic and foreign assets are perfect substitutes and investors can instantaneously and costlessly switch between the two asset types. In a small open economy which cannot affect the world interest rate or any other foreign macroeconomic variable, perfect capital mobility implies that the correlation between national saving and national investment tends to zero.
6. In our Solow model of a small open economy with perfect capital mobility, the domestic capital intensity adjusts instantaneously to its steady state level determined by the international interest rate. By contrast, national wealth per capita evolves gradually over time, since any increase in wealth generates an increase in capital income, thereby raising aggregate saving which in turn generates another increase in wealth, and so on.
7. The process of wealth accumulation in the small open economy is dynamically stable if and only if the (effective) labour force grows at a rate exceeding the product of the savings rate and the international real interest rate. When the labour force is measured in technology-adjusted efficiency units, this stability condition will be met for all realistic parameter values.
8. By allowing free international capital flows, a country will be able to increase its national income whenever the domestic steady state real interest rate in the absence

of capital mobility, r_c^* , deviates from the world real interest rate, $\bar{r}$. If $r_c^* > \bar{r}$, which will typically be the case in a country with a relatively low savings rate, capital mobility benefits the domestic economy by enabling firms to borrow at a world interest rate below the (initial) marginal return to domestic investment. If $r_c^* < \bar{r}$, which will tend to be the case in a country with a relatively high savings rate, domestic wealth owners gain because capital mobility allows them to invest their savings at a higher (initial) rate of return abroad than at home. If wealth is unevenly distributed, capital mobility will affect the distribution of income and may in fact hurt some domestic agents even though it raises aggregate income.

9. While capital mobility enables domestic agents to take advantage of rate-of-return differentials between the domestic and the foreign economy, it also makes countries more vulnerable to international interest rate shocks. In particular, a permanent increase in the real interest rate at which a net debtor country can borrow abroad will reduce the country's steady state level of national income and wealth. Moreover, if such an increase in the country risk premium occurs abruptly, as is often the case, it may create a serious domestic recession in the short run.
10. A risk premium on investment in a small economy may explain why capital flows out of the country, not into it, upon liberalization of capital movements, even if the country initially had a relatively high real interest rate. This is probably the background for the Mexican experience after NAFTA (and other similar incidences). This represents a serious challenge to the more orthodox view that capital mobility should lead to better living conditions for the working poor in low-saving countries through capital imports.

4.7 Exercises

Exercise 1. The elasticity of long-run national income per capita with respect to the savings rate in the small open economy

This exercise investigates how strongly national income per capita is influenced by the savings rate in the open economy compared to the closed.

1. First show from (27) that the elasticity of y^{**} with respect to s is:

$$e_o = \frac{(s/n)\bar{r}}{1 - (s/n)\bar{r}}$$

(the stability condition, $(s/n)\bar{r} < 1$, should be assumed).

In the closed economy studied in Chapter 3, the elasticity of national income (equal to GDP) per capita with respect to the savings rate, s , was found to be $e_c = \alpha/(1 - \alpha)$. It seems difficult to compare these two elasticities since they involve different parameters. We can, however, apply a trick we used in this chapter as well. The real interest rate that would emerge in the long run in the small open economy if it were closed (given $\delta = 0$) as stated in (24) was:

$$r_c^* = \alpha n / s.$$

2. Show that the elasticity, e_o , can be written as:

$$e_o = \frac{\alpha \bar{r}}{1 - \alpha \bar{r}}, \quad \text{where } \bar{r} \equiv \frac{\bar{r}}{r_c^*},$$

where the stability condition is now $\alpha\bar{r} < 1$. Show that if the long-run real interest rate that the open economy would produce if it were closed is equal to the international real interest rate, then $e_o = e_c$. Then show that if the considered small economy has a stronger propensity to save (as measured by s/n) than corresponding to $r_c^* = \bar{r}$, then $e_o > e_c$, and vice versa.

3. Try to explain your result intuitively. (Hint: If the economy considered is 'savings strong', so that $r_c^* < \bar{r}$, then, when it has adjusted fully to the conditions of free capital movements it will have taken over a rate of return on wealth that is higher than the one it would produce itself if it were closed.)

Exercise 2. Golden rule in the open economy?

Show that in the small open economy's steady state, consumption per capita is:

$$c^* = (1-s) \frac{1}{1-(s/n)\bar{r}} w^*,$$

where the expression for w^* of (19) could be inserted (but, since w^* does not depend on s , this is not of importance for the present exercise). Show that as long as the stability condition, $(s/n)\bar{r} < 1$, is fulfilled, an increase in s has two opposing effects on c^* , and explain the origins of these.

Given the two opposing effects it is natural to ask if there is a specific (golden rule) value of s that maximizes c^* . Show that this is not the case by demonstrating that as s increases up towards $n/\bar{r}$ (which is realistically assumed to be smaller than one), c^* goes to infinity.

Try to explain intuitively what goes on here. Explain first, for instance, what happens to steady state national income when s increases towards $n/\bar{r}$. To understand this, in turn, explain what happens to the transition curve as s increases towards $n/\bar{r}$, and explain why the transition curve rotates as it does.

Exercise 3. The income distribution in the small open economy

As stated in this chapter, in any period t , the share of labour income in GDP, w_t/y_t , is equal to $1 - \alpha$, while the share of income earned on domestic capital in GDP, $\bar{r}k_t/y_t$, is α . These features are unchanged from the model for the closed economy of Chapter 3. However, from the point of view of the small open economy the income shares of the resources *owned* by domestic residents in *national income* (rather than GDP) are of more interest. That is, the income distribution is more relevantly described by the share of labour income in national income, w_t/y_t^n , and the share of income earned on national wealth in national income, $\bar{r}v_t/y_t^n$.

1. Show, using (16) and the fact that w_t jumps immediately to w^* , that labour's share in any period t is:

$$\frac{w_t}{y_t^n} = 1 - \bar{r} \frac{v_t}{y_t^n} = 1 - \bar{r} \frac{v_t}{w^* + \bar{r}v_t},$$

and show that as national wealth, v_t , converges towards its steady state level, v^* , from below (above), labour's share will converge to its steady state level from above (below).

2. Show that in steady state the shares in national income of labour and wealth, respectively, are:

$$\frac{w^*}{y^{n*}} = 1 - \frac{s}{n} \bar{r} \quad \text{and} \quad \frac{\bar{r}v^*}{y^{n*}} = \frac{s}{n} \bar{r}.$$

(As usual we assume the stability condition, $(s/n)\bar{r} < 1$.) Show that in an open economy, a relatively strong propensity to save (as measured by s/n) will tend to create a relatively low labour share in the long run. Will it also imply a lower long-run wage rate, w^* ?

Explain your results intuitively. Should wage earners in open economies try to resist national saving?

3. To compare the functional income distribution just found to the income distribution in the closed economy, one can (once again) use the trick of rewriting some expressions in terms of the interest rate that the small economy *would* have created in the long run if it were closed, $r_c^* = \alpha n/s$. Show that the steady state functional income distribution in the open economy is given by:

$$\frac{w^*}{y^{n*}} = 1 - \alpha \bar{r} \quad \text{and} \quad \frac{\bar{r} V^*}{y^{n*}} = \alpha \bar{r}, \quad \text{where } \bar{r} \equiv \frac{\bar{r}}{r_c^*},$$

where the stability condition, $\alpha \bar{r} < 1$, is assumed. Show that if an economy has a relatively high propensity to save by international standards, an opening of the economy will imply a lower labour share in the long run. Explain why. Will an opening also imply a lower long-run wage, w^* ? Do wage earners in savings strong economies have reasons to try to resist a liberalization of capital movements? Should they rather direct their political influence towards other aims?

4. How is the long-run functional income distribution in the open economy affected by a permanent international interest rate shock, where $\bar{r}$ increases permanently to a new and higher level? Explain your result.

Exercise 4. The effects of a permanent increase in the savings rate

Consider the Solow model of the small open economy and assume that as from some period the savings rate, s , shifts permanently upwards to a new and higher level, s' . The shift is sufficiently small to ensure that the stability condition is fulfilled after the change, and the country's position as net debtor or creditor in steady state (the sign of f^*) does not change.

1. Explain that this gives rise to a change in the transition curve that can be described as a combination of a parallel shift upwards and an anti-clockwise rotation around the intersection point with the v_{t-1} axis. Explain the economic background for each of these two movements.
2. Assume that the economy was in steady state before the shift in the savings rate. Using your transition diagram, explain qualitatively the economy's reactions over time to the permanent increase in the savings rate. Focus first on the first period with s' . How are GDP per capita, the wage rate, national income per capita, consumption per capita and savings per capita affected in that period? Then consider the subsequent sequence of periods. How do GDP per capita and the wage rate evolve over those periods? Describe how national wealth per capita evolves, that is, describe period by period the process of wealth accumulation. How do national income and consumption per capita evolve?
3. Compare the old to the new steady state. How (in what direction) are GDP per capita and national income per capita changed? Can we be sure in what direction consumption per capita has changed? How do the country's net foreign assets change from the old to the new steady state?

Exercise 5. The importance of capital movements for convergence and a permanent productivity shock in the open vs. the closed economy

In this chapter it was shown that if the small open economy, in the case where it was closed, were to create a long-run interest rate, $r_c^* = \alpha n/s$, equal to the international interest rate, $\bar{r}$, then its steady state national income per capita would be the same as if it were closed. One may get the impression that if $\alpha n/s = \bar{r}$, then it does not matter if the economy is closed or open. This is absolutely false. It is only the steady state that is unaffected. What happens outside

steady state depends very much on the economy being closed or open, as this exercise illustrates. We will consider one and the same small economy as open and as closed.

We assume throughout that indeed $r_c^* = \alpha n/s = \bar{r}$, and that $\delta = 0$.

1. Show from (27) that under these assumptions steady state national income per capita is:

$$y^{n*} = B^{\frac{1}{1-\alpha}} \left(\frac{s}{n} \right)^{\frac{\alpha}{1-\alpha}}.$$

Then show, by comparing it to the relevant formula in Chapter 3, that this y^{n*} is equal to the steady state GDP per capita, y_c^* , for the closed economy. Show further that steady state national wealth per capita, v^* , for the open economy is equal to the steady state capital intensity, k_c^* , for the closed economy.

2. Consider the transition equation for the open economy (v_{t+1} as a function of v_t) and show that under our assumptions its slope is:

$$\frac{1 + \alpha n}{1 + n}.$$

Then consider the transition equation for the closed economy that we found in Chapter 3 (k_{t+1} as a function of k_t) and show that *at steady state* (for $k_t = k_c^*$) its slope is $(1 + \alpha n)/(1 + n)$ as well. On the basis of your findings in Questions 1 and 2, illustrate (in principle) the transition curves by drawing both in the same diagram (so you have both v_t and k_t along the horizontal axis, etc.). Assume that v_t and k_t for the open and the closed economy, respectively, start at a common initial value, $v_0 = k_0$, below steady state. For which economy will convergence towards steady state be fastest, the closed or the open, do you think? Explain intuitively *why* the fastest one is the fastest.

We will now assume that (initially) $B = 1$, which is just a normalization, and for the other parameters (except δ which has already been set equal to zero) assume some plausible values (on an annual basis): $\alpha = 0.4$, $n = 0.01$, $s = 0.2$. This implies that $r_c^* = \alpha n/s = 0.02$, so we assume that the annual international real interest rate is 2 per cent.

3. Compute the steady state values, $y^{n*} = y_c^*$ and $v^* = k_c^*$, under these specifications. Assume that both economies, the open and the closed, start in their steady states and then, as from period one, say, the total factor productivity increases permanently up to $B' = 2$. Compute the new steady state values for national income per capita and for wealth per capita for both economies. Describe in principle how the permanent productivity shock considered affects the transition diagrams and the accumulation of wealth and capital, respectively, in the open and the closed economy.
4. Now simulate each of the transition equations from period zero onwards (for a number of periods of your own choice, but be sure to have quite a few), thereby creating a sequence $\{v_t\}$ for the open economy and a sequence $\{k_t\}$ for the closed. From these sequences compute the associated sequences, $\{y_t^n\}$ and $\{y_t\}$, respectively, for national income per capita. Illustrate the output of your simulations by plotting each of y_t^n and y_t against periods in one and the same diagram that also has the old and the new steady states indicated by horizontal lines. Explain in words the differences and similarities between the series $\{y_t^n\}$ and $\{y_t\}$.

Exercise 6. Does the empirical evidence indicate that capital mobility is not important?

In this chapter we found that for a (small) economy with free capital movements, long-run GDP (not national income) per capita, y^* , is independent of s and n ; see (18). In Fig. 3.7 we found a relatively clear positive correlation between average investment shares and GDP per capita across countries, and in Fig. 3.8 we found a negative correlation between average population growth rates and GDP per capita. The question we address in this exercise is

whether we should conclude from the empirical evidence presented in Figs 3.7 and 3.8 that real-world economies behave more as closed than as open economies and that capital movements therefore are not so important for observed economic performance.

First, note that the variable indicated along the horizontal axis in Fig. 3.7 was *investment* shares of GDP, not *savings* shares. In a closed economy these coincide, but in an open economy they do not. Show, using (18) and (21), that for a small open economy in steady state there is the following necessary link between GDP per capita, y^* , the investment rate, i^* , and the growth rate of the labour force, n :

$$\ln y^* = \frac{1}{1-\alpha} \ln B + \frac{\alpha}{1-\alpha} (\ln i^* - \ln n).$$

Compare this to the expression for $\ln y^*$ we found in Section 3.3, and discuss whether the empirical evidence reported in Figs 3.7 and 3.8 indicates that capital movements are not important for economic performance across countries.

Exercise 7. Long-run national income in the small open economy as depending on domestic and international savings propensities

A certain restatement of (27) can be of interest. As argued in this chapter, the world economy is a closed system, so it should behave according to the model of Chapter 3 in the aggregate. This means that the formulae in Section 3.3 should be valid for the steady state of the world economy. If we assume that technology is the same in the domestic country and in the world (α is the same, and $\delta = 0$ in both), and we denote the average savings rate in the world by $\bar{s}$, and the world population growth rate by $\bar{n}$, then the steady state value for the world real interest rate is:

$$\bar{r} = \frac{\alpha \bar{n}}{\bar{s}}.$$

The meaning of the economy considered being *small* is exactly that, although it contributes to world savings and population growth, its impacts on $\bar{s}$ and $\bar{n}$ are so small that in practice $\bar{r}$ is given.⁹

Show, using (27), that steady state national income per capita is:

$$y^{n*} = (1-\alpha)B^{\frac{1}{1-\alpha}} \left(\frac{\bar{s}}{\bar{n}} \right)^{\frac{\alpha}{1-\alpha}} \cdot \frac{1}{1-\alpha \frac{s/n}{\bar{s}/\bar{n}}}.$$

Show that in the particular case, $s/n = \bar{s}/\bar{n}$, national income per capita is the same in the open economy as it would be if the economy were closed. What if $s/n > \bar{s}/\bar{n}$? Explain.

Exercise 8. The growth perspectives for a relatively poor, emerging market economy opening its capital account

This exercise considers a small economy that is relatively poor by world standards due to a relatively low savings capacity (as measured by s/n).

Our interest is in what the relatively poor economy might achieve, if anything, from opening its capital account, in particular the speed by which the effects occur.

Assuming a period length of one year, the economy has the following structural characteristics: $B = 1$, $\alpha = 1/3$, $\delta = 0$, $s = 1/6$, $n = 0.03$. The international real interest rate is $\bar{r} = 0.03$.

9. Note that if n is larger than $\bar{n}$, then the domestic economy will grow faster than the world economy, and at some point our assumption that the domestic economy is too small to affect the world economy will no longer be valid. We assume that this happens too far into the future to be of practical relevance.

1. The economy considered has a savings capacity, s/n , of around five and a half. Consult Table A to see if this seems reasonably descriptive for a typical poor economy.
First assume that the small economy does not allow capital movements between itself and the world economy.
2. Describe the long-run economic outcome according to the basic Solow model of the closed economy in terms of the key macroeconomic variables: capital and GDP (income) per capita, consumption and investment per capita, real interest rate and real wage rate.
Now assume that the small economy up to and including a specific year zero is in steady state as a closed economy, but liberalizes capital movements completely from period one and onwards, and thus comes to work as a small open economy (first with no country risk).
3. Is the stability condition fulfilled for the small and now open economy?
4. Describe the long-run (steady state) economic outcome of the small open economy in terms of all the variables mentioned in Question 2 and, in addition, in terms of national wealth, national income, net foreign assets, and interest payments on net foreign assets, all in per capita terms. Based on a comparison between the long-run outcomes of the small economy under a closed capital account and an open capital account, respectively, comment on the long-run perspectives for the small economy of opening up.
5. In this question, focus completely on period one, the first period of an open capital account for the small economy considered. For this period, state the values of all the variables that were mentioned in Question 4. (Be careful here: when you compute v_1 , you cannot just apply formula (22) of this chapter with $t + 1$ corresponding to one, and t corresponding to zero, because this formula assumes that the economy was open in period t . You can, however, use the formula four lines above.)
6. By simulating the appropriate model, create a table that reports the evolution over time of the central macroeconomic variables asked about in Questions 4 and 5 and, in addition, net capital inflow per capita and the growth rates of GDP per capita and of national income per capita, over years 0, 1, 2, ... up to 25. Illustrate in (at least) two figures with time in years along the horizontal axis. One should have wealth per capita along the vertical axis, and another should have GDP per capita, national income per capita, and real interest payments per capita, respectively, along the vertical axis. Describe the economy's process of convergence towards steady state and comment on what you think the analysis importantly says about the short- and intermediate-run 'growth perspectives for a relatively poor, emerging market economy of opening its capital account'.
As discussed by the end of this chapter, it is an important empirical observation that sometimes capital to a surprisingly small extent moves into, and sometimes out of, poor economies when they liberalize capital movements.
7. Set up a model for the small, open economy that includes country risk, assuming that the country is a net debtor. Assume an appropriate value for the country risk parameter, and show by doing relevant computations and simulations that, as a consequence of country risk, capital may move into the small economy to a (much) smaller extent after a liberalization of capital movements than found above. Comment on your results. Could country risk even imply that capital moves out of the small economy upon liberalization? How large would the country risk parameter have to be, and what would the appropriate model look like? (Don't analyse it, just describe it. Analysis comes in the next exercise.)

Exercise 9. Country risk and a risk premium shock if the small open economy is a net creditor

We argued in the chapter that to take country risk into account for a net creditor one should assume that the country earns the international interest rate without a risk premium on its net foreign assets since these are placed abroad. On the other hand the 'hurdle' rate of

return on domestic capital that will make investors consider wealth placed in the small open economy and in the international capital market equally good should include a risk premium. The appropriate model is therefore (7)–(14), with the $\bar{r}$ in (9) kept unchanged, and with the $\bar{r}$ in (13) replaced by $\bar{r} + \varepsilon$. We will consider the most standard case of $\varepsilon > 0$, and for simplicity assume $B = 1$. Otherwise notation is as in the chapter.

1. Write down the complete model (for your convenience) and show that in any period t , GDP per capita is:

$$y_t = w_t + (\bar{r} + \varepsilon)k_t,$$

while national income per capita is:

$$y_t^n = w_t + \varepsilon k_t + \bar{r}v_t.$$

2. Show that domestic capital per capita and the wage rate, respectively, adjust immediately to:

$$k^* = \left(\frac{\alpha}{\bar{r} + \varepsilon} \right)^{\frac{1}{1-\alpha}} \quad \text{and} \quad w^* = (1 - \alpha) \left(\frac{\alpha}{\bar{r} + \varepsilon} \right)^{\frac{\alpha}{1-\alpha}}.$$

3. Show that the transition equation for national wealth per capita is:

$$v_{t+1} = \frac{1 + s\bar{r}}{1 + n} v_t + \frac{s(w^* + \varepsilon k^*)}{1 + n}.$$

4. Using the above expressions for k^* and w^* , show that:

$$w^* + \varepsilon k^* = \left[1 + \frac{\varepsilon}{\bar{r} + \varepsilon} \frac{\alpha}{1 - \alpha} \right] (1 - \alpha) \left(\frac{\alpha}{\bar{r} + \varepsilon} \right)^{\frac{\alpha}{1-\alpha}}.$$

5. Show that an increase in ε implies a decrease in $w^* + \varepsilon k^*$, whenever $\varepsilon > 0$ initially. How is the transition equation affected by an increase in ε ?
6. Show that the steady state values for GDP per capita, the wage rate, wealth per capita and national income per capita all decrease in response to the increase in ε . (Hint: You don't have to solve the model for y^{n*} . From the shift of the transition curve you can infer how steady state wealth per capita, v^* , is affected. You can then use $y_t^n = w_t + \varepsilon k_t + \bar{r}v_t$, etc., to infer how steady state national income per capita, y^{n*} , is affected.) Explain your results intuitively.

Exercise 10. Free capital movements: pros and cons

On the background of what you have learned in this chapter, discuss arguments for and against liberalizing capital flows. Is your overall conclusion based on the theory (and empirics) of this chapter in favour of or against such liberalization?

PART 2 Exogenous Growth

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